

Physical Chemistry of Materials

P05:

Magnetic properties of materials

Magnetic terms.^[Mazzoldi, Strobl]

In the treatment of magnetism in a material three main quantities can be identified:¹

H : the magnetizing field, i.e. applied from the outside

B : magnetic induction, i.e. the magnetic field present inside the material

M : magnetization.

Magnetization M represents the density of magnetic moments per unit volume in a material. In analogy to polarization P , which in general (but not always) develops by application of an external electric field, magnetization in general (but not always) also results from the application of an external magnetic field H . Therefore, M will be proportional to the applied field with a coefficient of proportionality equal to the magnetic susceptibility χ_m .

$$\bar{M} = \chi_m \bar{H}$$

Since M and H are two tensors of 1st rank, χ_m will be a symmetric 2nd rank tensor.²

By analogy to the relative dielectric constant $\epsilon_r = \epsilon/\epsilon_0$ for dielectric properties, relative magnetic permeability (μ_r) is used to describe the magnetic properties of a material.

$$\bar{B} = \underline{\mu} \bar{H} = \mu_0 \underline{\mu_r} \bar{H}$$

where³

$$\underline{\mu_r} = \frac{\underline{\mu}}{\mu_0}$$

The relationship between $\underline{\mu_r}$ and χ_m is

$$\underline{\mu_r} = 1 + \underline{\chi_m} \quad (\text{S.I.})$$

whereby

$$\bar{B} = (1 + \underline{\chi_m}) \mu_0 \bar{H} \quad (\text{S.I.})$$

¹ In analogy to what will be seen in P06 for dielectric properties by introducing E , D and P .

² Again, χ_m symmetry derives for the same reasons that will be shown in P06 about electrical susceptibility χ_e and ϵ .

³ The unit of measurement of magnetic permeability is H/m , where it is recalled that the Henry $H = Vs/A$ is the unit of measurement used for electrical inductance.

So there are four ways in which the resulting magnetic induction B in a material can be defined:⁴

$$\underline{\bar{B}} = \underline{\mu} \underline{\bar{H}}$$

$$\underline{\bar{B}} = \underline{\mu_r} \underline{\mu_0} \underline{\bar{H}}$$

$$\underline{\bar{B}} = \underline{\mu_0} \underline{\bar{H}} + \underline{\mu_0} \underline{\bar{M}}$$

$$\underline{\bar{B}} = (1 + \underline{\chi_m}) \underline{\mu_0} \underline{\bar{H}}$$

The units of measurement of the physical quantities of interest for magnetism are shown in the following table:

Magnetic Term	Symbol	SI unit	CGS unit	conversion factor
magnetic induction	B	Tesla (T)	Gauss (G)	1 T = 10 ⁴ G
magnetic field	H	A/m	Oersted (Oe)	1 A/m = 4π/10 ³ Oe
magnetization	M	A/m	emu/cm ³	1 A/m = 10 ⁻³ emu /cm ³
mass magnetization	σ	Am ² /kg	emu/g	1 Am ² /kg = 1 emu/g
magnetic moment	m	Am ²	emu	1 Am ² = 10 ³ emu
volume susceptibility	κ	dimensionless	dimensionless	4π(SI) = 1 (cgs)
mass susceptibility	χ	m ³ /kg	emu/Oe·g	1 m ³ /kg = 10 ³ /4π emu /Oe·g
permeability of free space	μ ₀	H/m	dimensionless	4πx10 ⁻⁷ H/m = 1 (cgs)

⁴ Sometimes, the units of measurement in the S.I. system are preferred to those c.g.s., which, however, are still found in old texts and in scientific works not harmonized with the trend most shared internationally.

UNITS FOR MAGNETIC PROPERTIES

Quantity	Symbol	Gaussian & cgs emu ^a	Conversion factor, C ^b	SI & rationalized mks ^c
Magnetic flux density, magnetic induction	B	gauss (G) ^d	10^{-4}	tesla (T), Wb/m ²
Magnetic flux	Φ	maxwell (Mx), G·cm ²	10^{-8}	weber (Wb), volt second (V·s)
Magnetic potential difference, magnetomotive force	U, F	gilbert (Gb)	$10/4\pi$	ampere (A)
Magnetic field strength, magnetizing force	H	oersted (Oe), ^e Gb/cm	$10^3/4\pi$	A/m ^f
(Volume) magnetization ^g	M	emu/cm ³ ^h	10^3	A/m
(Volume) magnetization	$4\pi M$	G	$10^3/4\pi$	A/m
Magnetic polarization, intensity of magnetization	J, I	emu/cm ³	$4\pi \times 10^{-4}$	T, Wb/m ² ⁱ
(Mass) magnetization	σ, M	emu/g	$\frac{1}{4\pi \times 10^{-7}}$	A·m ² /kg Wb·m/kg
Magnetic moment	m	emu, erg/G	10^{-3}	A·m ² , joule per tesla (J/T)
Magnetic dipole moment	j	emu, erg/G	$4\pi \times 10^{-10}$	Wb·m ⁱ
(Volume) susceptibility	χ, κ	dimensionless, emu/cm ³	$\frac{4\pi}{(4\pi)^2 \times 10^{-7}}$	dimensionless henry per meter (H/m), Wb/(A·m)
(Mass) susceptibility	χ_p, κ_p	cm ³ /g, emu/g	$\frac{4\pi \times 10^{-3}}{(4\pi)^2 \times 10^{-10}}$	m ³ /kg H·m ² /kg
(Molar) susceptibility	$\chi_{\text{mol}}, \kappa_{\text{mol}}$	cm ³ /mol, emu/mol	$\frac{4\pi \times 10^{-6}}{(4\pi)^2 \times 10^{-13}}$	m ³ /mol H·m ² /mol
Permeability	μ	dimensionless	$4\pi \times 10^{-7}$	H/m, Wb/(A·m)
Relative permeability ^j	μ_r	not defined		dimensionless
(Volume) energy density, energy product ^k	W	erg/cm ³	10^{-1}	J/m ³
Demagnetization factor	D, N	dimensionless	$1/4\pi$	dimensionless

a. Gaussian units and cgs emu are the same for magnetic properties. The defining relation is $B = H + 4\pi M$.

b. Multiply a number in Gaussian units by C to convert it to SI (e.g., $1 \text{ G} \times 10^{-4} \text{ T/G} = 10^{-4} \text{ T}$).

c. SI (*Système International d'Unités*) has been adopted by the National Bureau of Standards. Where two conversion factors are given, the upper one is recognized under, or consistent with, SI and is based on the definition $B = \mu_0(H + M)$, where $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$. The lower one is not recognized under SI and is based on the definition $B = \mu_0 H + J$, where the symbol I is often used in place of J .

d. $1 \text{ gauss} = 10^3 \text{ gamma } (\gamma)$.

e. Both oersted and gauss are expressed as $\text{cm}^{-1/2} \cdot \text{g}^{1/2} \cdot \text{s}^{-1}$ in terms of base units.

f. A/m was often expressed as "ampere-turn per meter" when used for magnetic field strength.

g. Magnetic moment per unit volume.

h. The designation "emu" is not a unit.

i. Recognized under SI, even though based on the definition $B = \mu_0 H + J$. See footnote c.

j. $\mu_r = \mu/\mu_0 = 1 + \chi$, all in SI. μ_r is equal to Gaussian μ .

k. $B \cdot H$ and $\mu_0 M \cdot H$ have SI units J/m³; $M \cdot H$ and $B \cdot H/4\pi$ have Gaussian units erg/cm³.

R. B. Goldfarb and F. R. Fickett, U.S. Department of Commerce, National Bureau of Standards, Boulder, Colorado 80303, March 1985
NBS Special Publication 696 For sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402

Note: units named after people should be capitalized, although this is not the case in this table.

Magnetism is a phenomenon that requires quantum mechanics to be described, as a classical system would have no magnetic properties, even if immersed in a magnetizing field.

At the atomic or molecular level, the magnetic properties of a material can be traced back to three sources:⁵

- spin (electronic or nuclear)
- the electron orbital angular momentum
- the variation of the electronic orbital angular momentum due to the effect of the magnetizing field.

Each angular momentum is associated with a magnetic moment $\bar{\mu}$ proportional to it. For example, given the spin angular momentum ($\bar{s}$) for an isolated electron, the orbital angular momentum ($\bar{l}$) for an isolated electron⁶, and the spin angular momentum ($\bar{I}$) for a nucleus, one can write:

$$\bar{\mu}_s \propto \bar{s}$$

$$\bar{\mu}_l \propto \bar{l}$$

$$\bar{\mu}_I \propto \bar{I}$$

The constant of proportionality is called the gyromagnetic ratio (γ) and is proportional to the inverse of the mass of the object with angular momentum:

γ electronic (γ_e):

$$\gamma_e = -\frac{e}{2m_e}$$

γ nuclear:

$$\gamma = g_I \frac{e}{2M_p}$$

where the nuclear g factor (g_I) ranges from about 6 to about -6 and is tabulated for each nucleus.

So for an isolated electron you have $\gamma_e \propto 1/m_e$ and for a proton you have $\gamma \propto 1/M_p$. In the case of the magnetic moment associated with the spin angular momentum, a corrective factor g (called "g-factor"⁷) is required with respect to the case of orbital angular momentum:

$$\bar{\mu}^{spin} = g\gamma_e \bar{s}$$

$$\bar{\mu}^{orbital} = \gamma_e \bar{l}$$

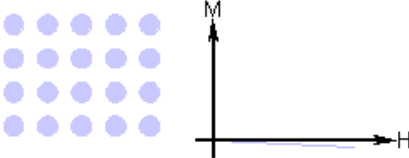
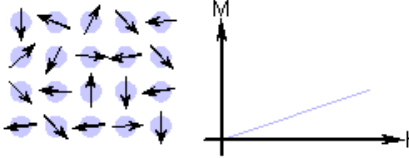
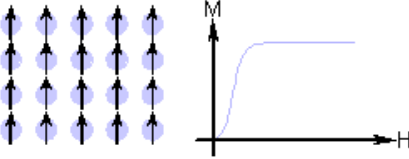
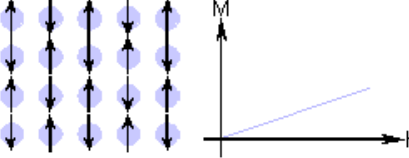
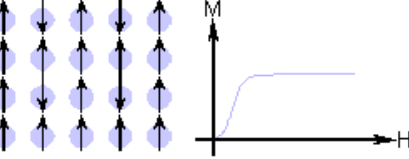
$$\bar{\mu}^{atomic} = \gamma \bar{I}$$

In matter, these three sources of magnetism mean that the magnetic behavior of materials can be divided into the following 5 categories:

⁵ Nuclear magnetic moments are about 10^{-3} times less intense than electronic ones.

⁶ For electrons in atoms, capital letters (S, L) will be used instead.

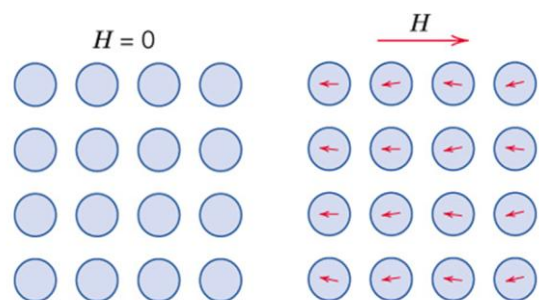
⁷ g is 2 for a free electron and 1 for an electron with zero spin and nonzero orbital angular moment, see Landè eq. later.

Type of Magnetism	Susceptibility	Atomic / Magnetic Behaviour		Example / Susceptibility	
Diamagnetism	Small & negative.	Atoms have no magnetic moment		Au Cu	-2.74×10^{-6} -0.77×10^{-6}
Paramagnetism	Small & positive.	Atoms have randomly oriented magnetic moments		β -Sn Pt Mn	0.19×10^{-6} 21.04×10^{-6} 66.10×10^{-6}
Ferromagnetism	Large & positive, function of applied field, microstructure dependent.	Atoms have parallel aligned magnetic moments		Fe	$\sim 100,000$
Antiferromagnetism	Small & positive.	Atoms have mixed parallel and anti-parallel aligned magnetic moments		Cr	3.6×10^{-6}
Ferrimagnetism	Large & positive, function of applied field, microstructure dependent	Atoms have anti-parallel aligned magnetic moments		Ba ferrite	~ 3

Diamagnetism^[Strobl]

Materials defined as diamagnetic are essentially "non-magnetic", in the sense that they do not exhibit magnetic behavior relevant for practical applications, and diamagnetism takes place only in the presence of an external magnetic field. In addition, the diamagnetic response in nature is extremely weak in intensity, compared to the magnitude of the magnetizing field.

In general, magnetism originates from electric currents (from electric charges subject to acceleration), and this applies both to "macroscopic" currents within conducting materials, and to "microscopic" currents associated with electron motion in individual atoms. In most cases, the magnetism due to "microscopic" currents is self-compensating and does not generate any effect outside the atom. However, this situation changes if a magnetic field is applied from the outside, because it induces a distortion of the electronic distribution at the atomic level, such that a magnetic



response is observed by the material even at the macroscopic level, due to the spontaneous tendency of the materials to oppose the external magnetizing field.

This phenomenon, known as diamagnetism, therefore consists in the appearance of an atomic magnetic moment in the opposite direction to the external field, and only in the presence of it.

Quantitatively, in diamagnets the relative magnetic permeability is just below 1

$$\mu_r = \frac{\mu}{\mu_0} \approx 0.99999 \rightarrow \chi_m = \mu_r - 1 \approx -10^{-5}$$

It should be noted that any material generates a diamagnetic response in the presence of a magnetizing field, however this response is relevant only when other forms of magnetism (paramagnetism, ferromagnetism, etc.) are not present, given its weak intensity.

This condition occurs in particular when the atoms that make up the material have completely full orbitals (that is, they are devoid of unpaired electrons to which it is possible to associate a net spin moment). Typical examples of these materials are:

- inert gases
- certain molecules (H₂O, organic molecules)
- Certain oxides and ionic solids (Al₂O₃)
- Noble metals (Au, Cu, Ag) and some transition metals (Hg, Cd, Zn, etc.)⁸
- graphite
- frogs...

Langevin's model for diamagnetism.^[Kittel, Strobl]

Although diamagnetism is a purely quantum mechanical effect, it can also be explained quantitatively by Langevin's semiclassical model.

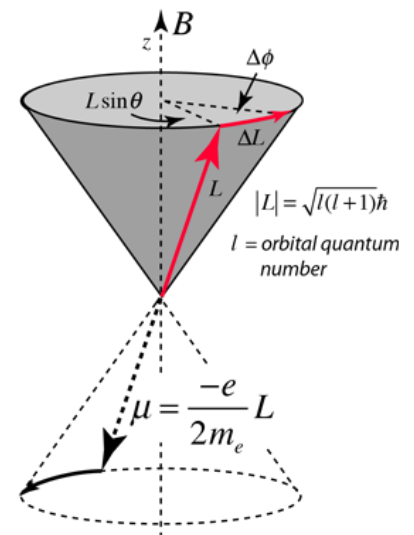
According to Larmor's theorem, the electrons of an atom immersed in a magnetic field B are subject to a precession with frequency

$$\omega_L = \frac{eB}{2m} \text{ (S.I.)}$$

which, according to the principle of magnetic induction⁹, involves the existence of a current called the Larmor precession current

$$I = (-Ze) \left(\frac{1}{2\pi} \frac{eB}{2m} \right) \text{ (S.I.)}$$

which is equivalent to the product of the charges concerned for the frequency of revolution.



⁸ The electronic configurations are [Hg] = [Xe] 4f¹⁴ 5d¹⁰ 6s², [Cd] = [Kr] 4d¹⁰ 5s² and [Zn] = [Ar] 3d¹⁰ 4s².

⁹ The principle of magnetic induction states that, if a magnetic field is applied to a conducting circuit, a "transient" electric current develops with a direction determined by Lenz's law, so as to oppose the external magnetic field.

In this case of an electron orbiting the nucleus, the current is not transient as predicted by the principle of magnetic induction according to classical electromagnetism, but persists as long as the field is present.

The magnetic moment associated with this current is obtained by multiplying the current itself by the area of the circuit on which the motion occurs. Assuming that the circuit has radius $\langle \rho \rangle$ (area $A = \pi \langle \rho^2 \rangle$):

$$\mu_a = IA = -\frac{Ze^2 B}{4m} \langle \rho^2 \rangle \text{ (S.I.)}$$

where m_e is the mass of the electrons, Z is the number of electrons in the atom, which are assumed to orbit all in the same circuit (hence why the average value of the circuit radius is used $\langle \rho \rangle$) and at the same Larmor frequency.

By the effect of Lenz's law, the direction of the magnetic moment will be such as to oppose the external field.

By convention the axis along which the magnetic field is located is identified with the z-axis, which will be the axis perpendicular to the surface A . So $\langle \rho \rangle$ it coincides with the average distance of the electrons from the axis of rotation:

$$\langle \rho \rangle = \langle r_{\perp} \rangle$$

Since atoms with complete electron shells, as is the case with diamagnetic elements, have charge distribution with spherical symmetry,

$$\langle x^2 \rangle = \langle y^2 \rangle = \langle z^2 \rangle$$

which allows you to write the average position of the electrons as

$$\langle r^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle + \langle z^2 \rangle = 3\langle x^2 \rangle = 3\langle y^2 \rangle = 3\langle z^2 \rangle$$

and the average distance of the electrons from the axis of rotation as

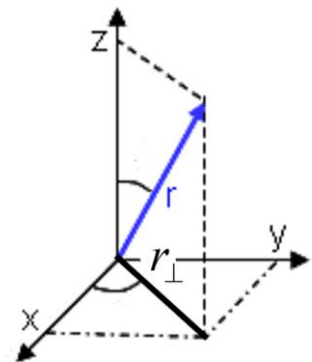
$$\langle \rho^2 \rangle = \langle r_{\perp}^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle = \frac{2}{3} \langle r^2 \rangle$$

from which

$$\mu_a = -\frac{Ze^2 B}{6m} \langle r^2 \rangle \text{ (S.I.)}$$

This leads to Langevin's law for magnetic susceptibility in diamagnetism:

$$\left. \begin{aligned} M &= N\mu_a \\ \chi_m &= \frac{M}{H} \approx \mu_0 \frac{M}{B} \end{aligned} \right\} \rightarrow \chi_{dia} = \mu_0 \frac{N\mu_a}{B} = -\mu_0 \frac{NZe^2}{6m} \langle r^2 \rangle \text{ (S.I.)}$$



where N is the number of atoms per unit volume, and use has been made of

$$B = \mu_0 H + \mu_0 M \approx \mu_0 H$$

since $\chi_m \ll 1 \rightarrow \mu_r \sim 1$ and M is very small.

The quantum mechanical theory introduces the interaction of the magnetizing field with the electrons as a small perturbation to the Hamiltonian of the system and, solving the problem with perturbation theory, arrives at the same result as Langevin's law.

The main limits of this model are related to the existence of closed electron shells, so in the case of metallic materials in which free electrons are present it is no longer valid, and to the hypothesis of spherical symmetry of the electron shell, for which in the case of molecules it is necessary to resort to an expression of χ_{dia} taking into account both the contribution of the electrons of the inner shells (χ_{inner}) and that of the valence shell participating in molecular bonds with non-spherical symmetry (χ_{bond}):

$$\chi_{dia} = \chi_{inner} + \chi_{bond}$$

Paramagnetism.^[Kittel]

For what has been said in the previous paragraphs, if at the atomic level, the electron orbitals are not completely full or the spins are not balanced, the result is a total non-zero angular momentum $\hbar \bar{J}$, or for which

$$|\hbar \bar{J}|^2 = \hbar^2 J(J+1) \neq 0 \rightarrow J \neq 0$$

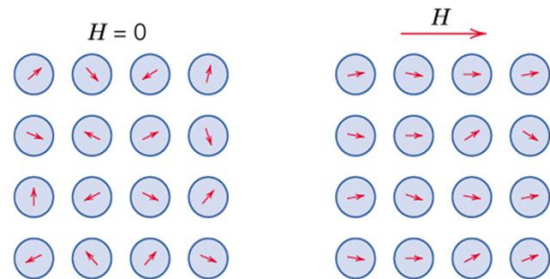
then there will be a permanent magnetic moment proportional to $\hbar \bar{J}$:

$$\bar{\mu}_a \propto \bar{J}$$

If it is assumed that there is no interaction between the microscopic magnetic moments, in the absence of an external magnetizing field the magnetic moments are randomly oriented, that is, the material has zero net magnetization and is "non-magnetic".

However, in an external field, the magnetic moments align with the field, amplifying it (albeit by a small amount compared to the magnetizing field itself).

Thus, the magnetic susceptibility of paramagnetic materials is positive ($\chi > 0$, $\mu_r > 1$) and the relative magnetic permeability typically varies between 1.00001 and 1.01.



Typically, molecules consist of atoms with closed electron shells, so J is zero, but there are exceptions where paramagnetic behavior is found:

- materials that contain atoms, molecules or point defects associated with the presence of an odd number of electrons;
- molecules with ground state of triplet (e.g. oxygen molecules, biradicals);
- materials containing atoms or ions with partially filled ¹⁰inner electron shells (e.g. some transition metals and related compounds such as Cr, Cr₂Cl₃, MnSO₄, Al), rare earths;
- some materials with conduction electrons such as metals.

Paramagnetism associated with spin and orbital angular momentums.

The magnetic moment of an atom or ion can be written as:

$$\bar{\mu}_a = g\gamma_e \bar{J} = -\frac{g\mu_B}{\hbar} \bar{J}$$

where

- γ_e is the gyromagnetic ratio
- The total angular momentum operator is given by:

$$\hbar \bar{J} = \hbar \bar{L} + \hbar \bar{S}$$

- μ_B is the Bohr magneton (m is the mass of the electron):

$$\mu_B = -\gamma_e \hbar = \frac{e\hbar}{2m_e} = 9.274 \cdot 10^{-24} \text{ J/T}$$

whose value is close to the magnetic spin moment of a free electron, for which $J = 1/2$ and $g = 2.0023$.

- g is the g -factor defined by the Landé equation:

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$$

It is observed that $g = 1$ if $J = L$ and $g = 2$ if $J = S$.

The Landé equation holds in cases of only "LS coupling" between L and S in an atom, i.e. when there are no external factors that influence the interaction between the two angular momentums, such as the presence of a "crystalline field" (see below) that cancels the contribution of orbital angular momentum, or the interaction with angular momentum, of other neighboring atoms. For example, the Landé equation applies very well to rare earths, whose paramagnetism is due to the presence of unpaired electrons in the f-shells, which are too internal to participate in the molecular bond and be affected by crystal field effects.

¹⁰ "Internal" means that they are not necessarily involved in the formation of molecular bonds leading to the formation of closed electron shells ($S = 0$, $L = 0$).

Curie's Law for Paramagnetism.^[Kittel]

In the presence of an external magnetic field $\vec{B}_0$, the "classical" energy of a magnetic moment is given by:

$$E = -\vec{\mu} \cdot \vec{B}_0$$

Similarly, the Hamiltonian of the quantum mechanical problem is:

$$\hat{H} = -\hat{\vec{\mu}} \cdot \vec{B}_0$$

I can therefore calculate the magnetic moment energy of the three examples seen above for isolated electron or generic nucleus in the presence of the magnetic field:

$$\vec{s} : E = -g\gamma_e \vec{s} \cdot \vec{B}_0 = -g\gamma_e \hbar m_s B_0$$

$$\vec{l} : E = -\gamma_e \vec{l} \cdot \vec{B}_0 = -\gamma_e \hbar m_l B_0$$

$$\vec{I} : E = -\gamma \vec{I} \cdot \vec{B}_0 = -\gamma \hbar m_I B_0$$

To simplify writing, the energy associated with electronic magnetic moments can be expressed as a function of the Bohr magneton μ_B :

$$\vec{s} : E = -g\gamma_e \vec{s} \cdot \vec{B}_0 = g\mu_B m_s B_0$$

$$\vec{l} : E = -\gamma_e \vec{l} \cdot \vec{B}_0 = \mu_B m_l B_0$$

Similarly, the energy associated with the magnetic moments of the nuclei can be expressed as a function of the nuclear magneton μ_N :

$$\mu_N = \frac{\gamma \hbar}{g_I} = \frac{e\hbar}{2M_p} = 5.051 \cdot 10^{-27} \text{ J/T}$$

$$\vec{I} : E = -\gamma \vec{I} \cdot \vec{B}_0 = -g_I \mu_N m_I B_0$$

In general, the energy associated with a magnetic moment in the presence of a magnetic field is

$$U = -\vec{\mu}_a \cdot \vec{B} = -\left(-\frac{g\mu_B}{\hbar} \vec{J}\right) \cdot \vec{B} = \frac{g\mu_B}{\hbar} \hbar m_J B = g\mu_B m_J B$$

where the projection of $\vec{J}$ along the axis of the magnetic field will be given by

$$J_z = \hbar m_J$$

with

$$m_J = J, (J-1), \dots, -(J-1), -J$$

Then the initially degenerate $2J+1$ levels are separated into energy by the magnetic field.

In the simple case where we have $J = 1/2$ (for example for a free electron with $S = 1/2$ and $L = 0$), we can identify only two states with populations N_1 ($m_J = -1/2$) and N_2 ($m_J = +1/2$) given by Boltzmann's statistics:

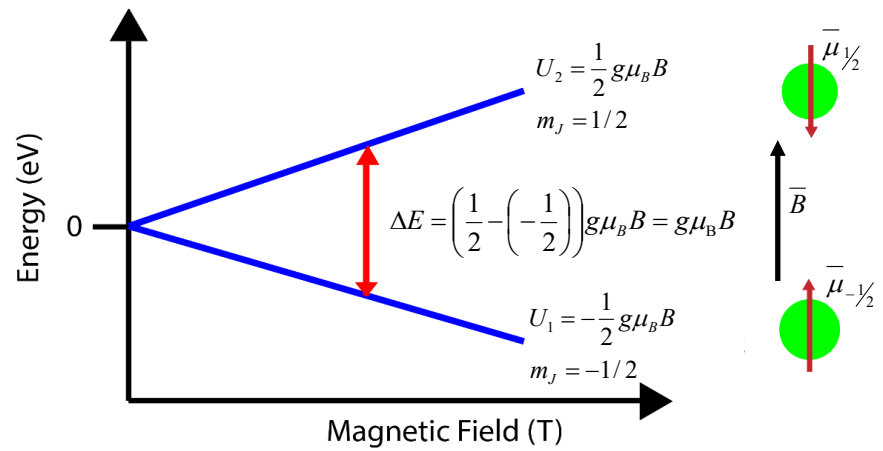
$$N_i = \frac{e^{-\frac{U_i}{kT}}}{e^{-\frac{U_1}{kT}} + e^{-\frac{U_2}{kT}}} N$$

with

$$U_1 = -\vec{\mu}_1 \cdot \vec{B} = -\frac{1}{2} g \mu_B B = -U_2$$

So, having

$$\mu_a = -g \frac{\mu_B}{\hbar} \hbar m_J = \frac{1}{2} g \mu_B$$



One can express magnetization as

$$\left. \begin{aligned} M &= (N_1 - N_2) \mu_a = N \mu_a \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ x &= \frac{\mu_a B}{k_B T} \end{aligned} \right\} \rightarrow M = N \mu_a \tanh(x)$$

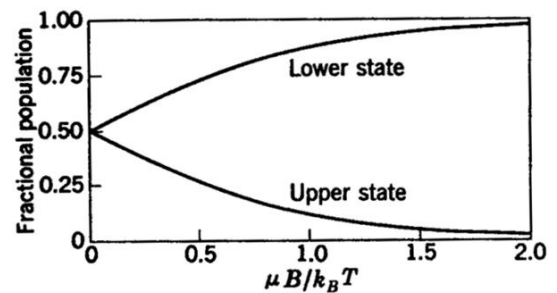
When $x \ll 1$ (i.e. at high temperature and/or for low external magnetic fields, since μ_a is always small), we have

$$\tanh(x) \approx x$$

therefore

$$M \approx \frac{N \mu_a^2 B}{k_B T}$$

As shown in the figure, the separation of the energy levels of an electron in the presence of an external magnetic field B increases with the modulus of B (by convention, the z axis is assumed parallel to B and with the same direction).



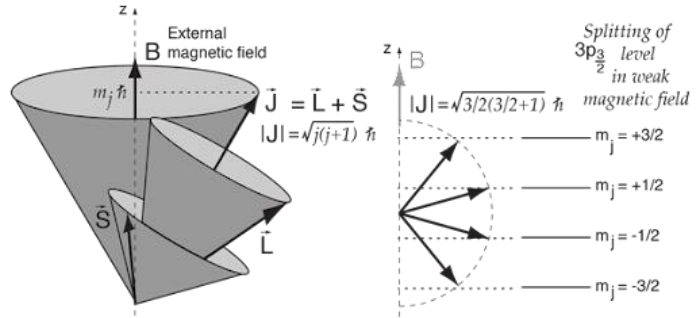
For an isolated electron, the magnetic moment μ_a has opposite sign to the spin $J (= S)$. Thus, in the lower energy state, the magnetic moment is parallel to the external field.

Such a separation in energy will correspond to a different population of the two levels, in thermal equilibrium at the temperature of the system. Macroscopically, the magnetization is proportional to the difference between the two curves.

In the case of a generic atom with a total angular momentum J , in the presence of an external magnetic field it will have $(2J + 1)$ equally spaced energy levels. One can then express the average magnetic moment using Boltzmann

statistics:

$$\langle \mu_a \rangle = \frac{\sum_{m_J=-J}^{-J} -m_J g \mu_B \exp\left(-\frac{g m_J \mu_B B}{k_B T}\right)}{\sum_{m_J=-J}^{-J} \exp\left(-\frac{g m_J \mu_B B}{k_B T}\right)}$$



This summation can be calculated exactly and leads to the Brillouin function $B_J(x)$:

$$B_J(x) = \frac{2J+1}{2J} \coth\left(\frac{(2J+1)x}{2J}\right) - \frac{1}{2J} \coth\left(\frac{x}{2J}\right)$$

such that

$$\langle \mu_a \rangle = g J \mu_B B_J(x)$$

$$x = \frac{g J \mu_B B}{k_B T}$$

So what

$$M = N g J \mu_B B_J(x)$$

In fact, for $S = 1/2$ we have

$$B_{J=\frac{1}{2}}(x) = \frac{1+1}{1} \coth\left(\frac{(1+1)x}{1}\right) - \frac{1}{1} \coth\left(\frac{x}{1}\right) = 2 \coth(2x) - \coth(x)$$

and being

$$\frac{1}{\tanh(x+x)} = \coth(x+x) = \frac{1 + \tanh(x)\tanh(x)}{\tanh(x) + \tanh(x)}$$

you have

$$B_{J=\frac{1}{2}}(x) = 2 \frac{1 + \tanh(x)\tanh(x)}{\tanh(x) + \tanh(x)} - \coth(x) = 2 \frac{1 + \tanh(x)^2}{2 \tanh(x)} - \frac{1}{\tanh(x)} = \frac{1 + \tanh(x)^2 - 1}{\tanh(x)} = \tanh(x)$$

Under ordinary conditions $x \ll 1$, and one can expand the $\coth$ in power series:

$$\coth(a) = \frac{1}{a} + \frac{a}{3} - \frac{a^3}{45} + \dots$$

Truncating the series expansion at the 2nd term, one arrives at:

$$\begin{aligned}
\langle \mu_a \rangle &= gJ\mu_B \left[\frac{2J+1}{2J} \coth\left(\frac{(2J+1)x}{2J}\right) - \frac{1}{2J} \coth\left(\frac{x}{2J}\right) \right] \approx \\
&\approx gJ\mu_B \left[\frac{2J+1}{2J} \left(\frac{2J}{(2J+1)x} \right) - \frac{1}{2J} \left(\frac{2J}{x} \right) + \frac{2J+1}{2J} \left(\frac{1}{3} \frac{(2J+1)x}{2J} \right) - \frac{1}{2J} \left(\frac{1}{3} \frac{x}{2J} \right) \right] = \\
&= gJ\mu_B \left[\frac{1}{3} \frac{(2J+1)^2}{(2J)^2} x - \frac{1}{3} \frac{1}{(2J)^2} x \right] = gJ\mu_B \left[\frac{1}{3} \frac{4J^2 + 4J + 1 - 1}{4J^2} x \right] = gJ\mu_B \frac{J+1}{3J} \frac{gJ\mu_B B}{k_B T} \\
&= J(J+1) \frac{g^2 \mu_B^2 B}{3k_B T}
\end{aligned}$$

which corresponds to a magnetization (N is always understood as the number of paramagnetic centers per unit volume):

$$M = N \langle \mu_a \rangle = NJ(J+1) \frac{g^2 \mu_B^2 B}{3k_B T}$$

If we consider that

$$B = \mu_0 H + \mu_0 M \approx \mu_0 H$$

since $\chi_m \ll 1 \rightarrow \mu_r \sim 1$ and M is very small, then one can write Curie's law for paramagnetism

$$\chi_{Curie} = \mu_0 \frac{M}{B} \approx \mu_0 NJ(J+1) \frac{g^2 \mu_B^2}{3k_B T} = \mu_0 \frac{Np^2 \mu_B^2}{3k_B T} = \frac{C}{T}$$

where p is called the effective number of Bohr magnetons:

$$p = g[J(J+1)]^{1/2}$$

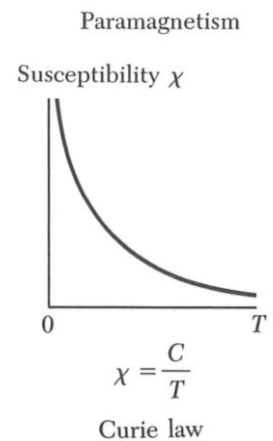
Since μ_B corresponds to the magnetic moment of a free electron, the meaning of p is to indicate the magnitude of the actual atomic magnetic moment in units μ_B .

C is Curie 's constant of the material:

$$C = \mu_0 \frac{Np^2 \mu_B^2}{3k_B} \quad (\text{S.I.})$$

$$C = \frac{Np^2 \mu_B^2}{3k_B} \quad (\text{C.G.S.})$$

and determines the magnitude of the paramagnetic response, depending on the density of paramagnetic centers for the square of their effective number of Bohr magnetons.



**Effective Magnetron Numbers p for Trivalent Lanthanide Group Ions
(Near room temperature)**

Ion	Configuration	Ground level	$p_{calc} = g[J(J+1)]^{1/2}$	p_{exp}
Ce ³⁺	4f ¹ 5s ² p ⁶	³ F _{5/2}	2.54	2.4
Pr ³⁺	4f ² 5s ² p ⁶	³ H ₄	3.58	3.5
Nd ³⁺	4f ³ 5s ² p ⁶	⁴ I _{9/2}	3.62	3.5
Pm ³⁺	4f ⁴ 5s ² p ⁶	⁵ I ₄	2.68	-
Sm ³⁺	4f ⁶ 5s ² p ⁶	⁶ H _{5/2}	0.84	1.5
Eu ³⁺	4f ⁶ 5s ² p ⁶	⁷ F ₀	0	3.4
Gd ³⁺	4f ⁷ 5s ² p ⁶	⁸ S _{7/2}	7.94	8.0
Tb ³⁺	4f ⁸ 5s ² p ⁶	⁷ F ₆	9.72	9.5
Dy ³⁺	4f ⁹ 5s ² p ⁶	⁶ H _{15/2}	10.63	10.6
Ho ³⁺	4f ¹⁰ 5s ² p ⁶	⁵ I ₈	10.60	10.4
Er ³⁺	4f ¹¹ 5s ² p ⁶	⁴ I _{15/2}	9.59	9.5
Tm ³⁺	4f ¹² 5s ² p ⁶	³ H ₆	7.57	7.3
Yb ³⁺	4f ¹³ 5s ² p ⁶	² F _{7/2}	4.54	4.5

The trivalent ions of the lanthanides¹¹ are distinguished by an electron configuration equivalent to that of [Xe] plus a variable number of electrons in the 4f orbitals. The ionic radius varies very little, having a maximum for Ce³⁺ at 1.11 Å and a minimum at 0.94 Å for Yb³⁺ (known as "lanthanide contraction"), due to the fact that the 4f shell is quite internal and generally not affected by the formation of chemical bonds.

Dy³⁺, Ho³⁺ and Er³⁺ are the lanthanides with the most intense paramagnetic response, but the order of magnitude is the same for everyone.

At room temperature, for all lanthanides, except Eu³⁺ and Sm³⁺, the agreement between the p obtained experimentally by applying Curie's law and the theoretical one is very good. For Eu³⁺ and Sm³⁺ the disagreement is due to the population of states with spin J at higher energy than the ground state, which is possible because the difference in energy with the ground state is comparable with the thermal energy at room temperature (to which the measurements refer).¹²

¹¹ According to IUPAC, lanthanides, along with Sc and Y, make up the "rare earths" group. In the table, promethium lacks the experimental result as it is not found in nature on Earth (its spectral lines have been detected in the light of some stars).

¹² The presence of multiple spin configurations is due to the spin orbit coupling effects between L and S , which lead to a multiplet of J states. In Eu³⁺ and Sm³⁺ at room temperature, other states with J other different from that of the ground state are also populated.

The value of p in the case of transition metals of the 4th period is very different from the lanthanides:¹³ the theory fails to predict the correct value, except in the case in which it is assumed that the orbital angular momentum L is zero, so that only the spin contribution to the total atomic angular momentum remains. For the rest, the values of p are of the same order of magnitude as that of the lanthanides.

Effective Magnetron Numbers for Iron Period Ions

Ion	Configuration	Ground level	$p_{calc} = g[J(J+1)]^{1/2}$	$p_{calc} = 2[S(S+1)]^{1/2}$	p_{exp}
Ti ³⁺ , V ⁴⁺	3d ¹	² D _{3/2}	1.55	1.73	1.8
V ³⁺	3d ²	² F ₂	1.63	2.83	2.8
Cr ³⁺ , V ²⁺	3d ³	⁴ F _{3/2}	0.77	3.87	3.8
Mn ³⁺ , Cr ²⁺	3d ⁴	⁵ D ₀	0	4.90	4.9
Fe ³⁺ , Mn ²⁺	3d ⁵	⁶ S _{5/2}	5.92	5.92	5.9
Fe ²⁺	3d ⁶	⁵ D ₄	6.70	4.90	5.4
Co ²⁺	3d ⁷	⁴ F _{9/2}	6.63	3.87	4.8
Ni ²⁺	3d ⁸	³ F ₄	5.59	2.83	3.2
Cu ²⁺	3d ⁹	² D _{5/2}	3.55	1.73	1.9

Because of ions close to atoms, shell d is polarized and J is no longer a good quantum number. In fact, the $3d$ shell in the metals of the 4th group is not internal like the $4f$ in the lanthanides, and this causes the d orbitals to suffer from the intense electric field due to the surrounding ions in the crystal. This spatially inhomogeneous electric field is called the "crystalline field".

For example, in the case of an ionic crystal with orthorhombic symmetry, the solution of the Laplace equation

$$\nabla^2 \phi = 0$$

where ϕ is the electrostatic potential, leads to a variable electric field with the position in the cell as

$$E_{cf}(x, y, z) = -\nabla \phi(x, y, z) = Ax^2 + By^2 - (A + B)z^2$$

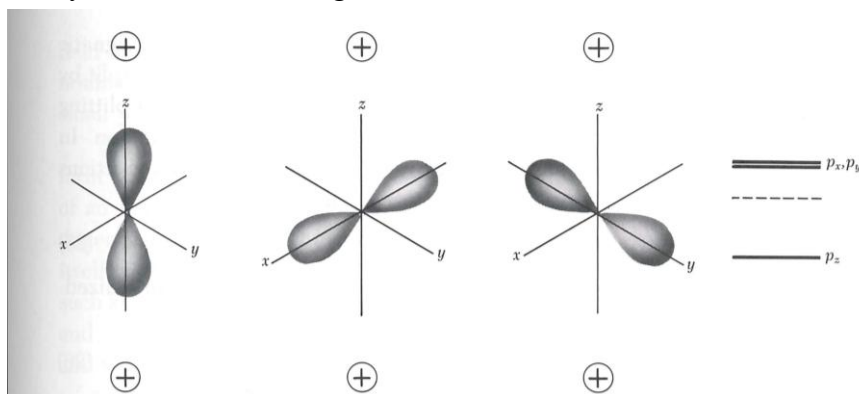
where A and B are two constants.

The crystalline electric field has two effects:

- the L - S coupling fails, so J is no longer a valid quantum number;
- the $2L+1$ levels associated with each L are no longer degenerate, but there is a splitting in energy. In fact, the interaction of the electron charge distribution of the $2L+1$ levels with the

¹³ The elements of the periodic table can be grouped by "period" or by "group". The 4th period is also called the iron period.

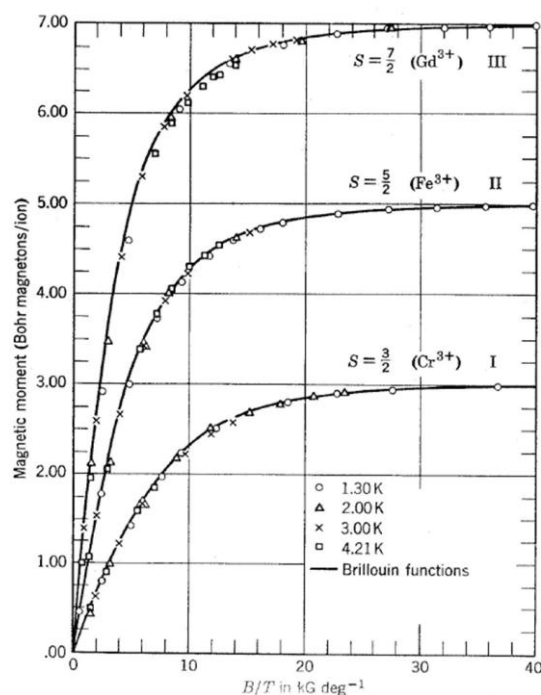
charges of the crystalline field (for example with cations or anions arranged along some specific directions), can be energetically more favorable or less favorable. The figure shows the hypothetical (simple) case of a shell with $L = 1$ in a uniaxial crystalline field due to the presence of two cations arranged along the z axis, so the p_z orbital will be energetically favored due to the greater proximity of the electric charge to the cations.



This splitting leads to a lower contribution of the orbital angular momentum to the overall magnetic moment of the atom, because inside a crystal the electric field is non-central, that is, it is not directed towards the center of the atom where the positive nucleus is located, and in this situation the projection of L along a certain direction z (L_z) is on average zero, that is, there is the "quenching" of angular momentum. In the presence of an external magnetic field along the z direction, the L_z component is zero and will remain the only spin angular momentum, if any, to contribute to the atomic magnetic moment μ_a .

The presence of spin orbit interactions between S and L leads to a slight modification of the g -factor compared to the case in which L is completely neglected, so $g > 2$ for full 3d shell over half, $g = 2$ for half-filled 3d shell, and $g < 2$ for 3d shell less than half full.

The comparison of the theoretical magnetization calculated by the expression leading to Curie's law, and the experimental results for lanthanide salts or transition metals indicates the validity of the theory, once the quenching of the orbital angular momentum is taken into account.¹⁴



¹⁴ The trend of the magnetic moment per magnetic center in the figure is linear for B/T small as per Curie's law, and then saturates when the ratio B/T increases, as expected by the complete expression as a function of the $\coth$, expected when the alignment of all the spins of the system is reached.

Pauli's paramagnetism.^[Kittel, Ziman]

Conduction electrons in metals are, to a first approximation, modelable as free electrons, each of which can be associated with a magnetic moment but $\neq \mu_B$. From this it could be assumed that metals have paramagnetic behavior, but experimentally it is verified that Curie's law is not applicable, as the magnetization of non-ferromagnetic metals is independent of temperature (see figure).

Pauli's theory describes the paramagnetic response of conduction electrons in a material and relies on the Fermi-Dirac distribution to correct Curie's theory, based on the fact that, in the conduction band of metals, the orientation of electronic magnetic moments along the magnetic field is not always possible because this would correspond to the occupation of already filled electronic levels. From other electrons of the same band. Only electrons in the levels near the Fermi E_F energy, within an energy range of the order of thermal quantum $k_B T$, will have the ability to switch from a more energetic to a less energetic spin configuration, aligned with the external magnetic field.

On the basis of this consideration, it can be inferred that only a fraction equal to T/T_F of the total number of conduction electrons N will contribute to susceptibility

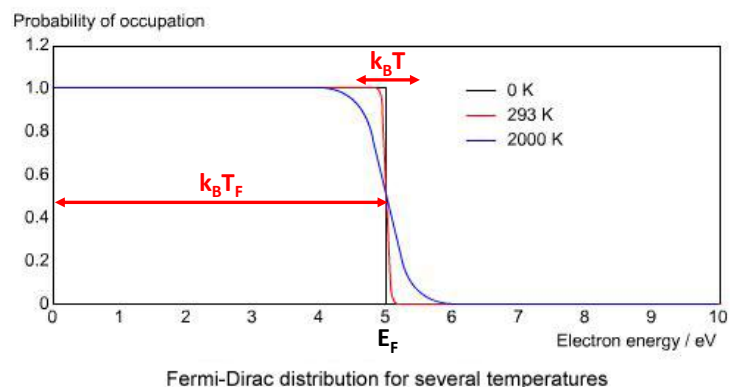
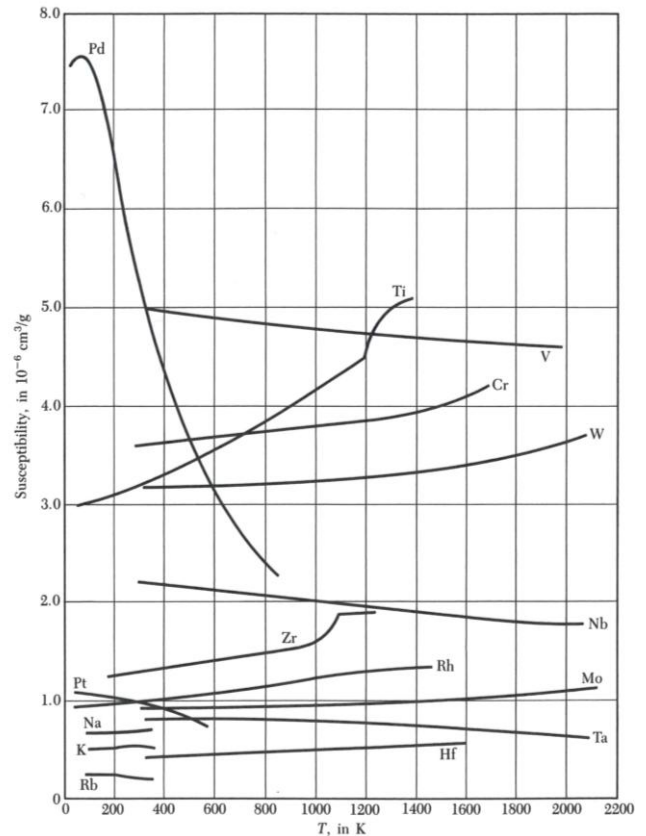
$$N' \approx \frac{k_B T}{k_B T_F} N$$

where

$$T_F = \frac{E_F}{k_B}$$

so the magnetization can be written as

$$\begin{aligned} M &= N' \langle \mu_a \rangle = N' J(J+1) \frac{g^2 \mu_B^2 B}{3k_B T} = \\ &= N' \frac{1}{2} \left(\frac{1}{2} + 1 \right) \frac{2^2 \mu_B^2 B}{3k_B T} = \end{aligned}$$



$$= \frac{N' \mu_a^2 B}{k_B T} \approx N \frac{\mu_a^2 B}{k_B T} \frac{T}{T_F} = \frac{N \mu_a^2 B}{k_B T_F}$$

which actually turns out to be independent of temperature.

A more rigorous expression of the Pauli magnetization can be derived if one assumes to be at absolute 0, and expresses the concentration of electrons with magnetic moment parallel to the magnetic field by the integral of the Fermi-Dirac distribution D :

$$N_{\uparrow} = \frac{1}{2} \int_{-\mu_a B}^{E_F} D(E + \mu_a B) dE$$

where the argument of the density of states D is increased by the energy of the magnetic moment in the presence of B , since at absolute 0 the occupation of the conduction band extends beyond E_F by a quantity $+\mu_a B$. In fact, as shown in the figure below, the electronic states with magnetic moment parallel to B will be energetically stabilized by a factor

$$U^{\uparrow} = -\bar{\mu}_a \cdot \bar{B} = -\frac{1}{2} g \mu_B B = -\mu_a B$$

while those with magnetic moment antiparallel to the field B will experience an increase in energy by a factor

$$U^{\downarrow} = -\bar{\mu}_a \cdot \bar{B} = +\frac{1}{2} g \mu_B B = +\mu_a B$$

although the Fermi level for the two electron subpopulations remains the same.

The factor 1/2 takes into account the fact that only half of the states at a given energy, those with magnetic moment parallel to the field, must be considered in the integral.

If one assumes that, in the vicinity of E_F , the density of the states is approximately uniform for an interval $\mu_a B$, then one can approximate $N_{\uparrow}$ as

$$N_{\uparrow} \approx \frac{1}{2} \int_0^{E_F} D(E) dE + \frac{1}{2} D(E_F) \mu_a B$$

The calculation of $N_{\downarrow}$ will be in analogy

$$N_{\downarrow} = \frac{1}{2} \int_{+\mu_a B}^{E_F} D(E - \mu_a B) dE \approx \frac{1}{2} \int_0^{E_F} D(E) dE - \frac{1}{2} D(E_F) \mu_a B$$

so one can express the magnetization as

$$\begin{aligned} M &= (N_{\uparrow} - N_{\downarrow}) \mu_a = \left(\frac{1}{2} \int_0^{E_F} D(E) dE + \frac{1}{2} \mu_a B D(E_F) - \frac{1}{2} \int_0^{E_F} D(E) dE + \frac{1}{2} \mu_a B D(E_F) \right) \mu_a \\ &= \mu_a^2 D(E_F) B \end{aligned}$$

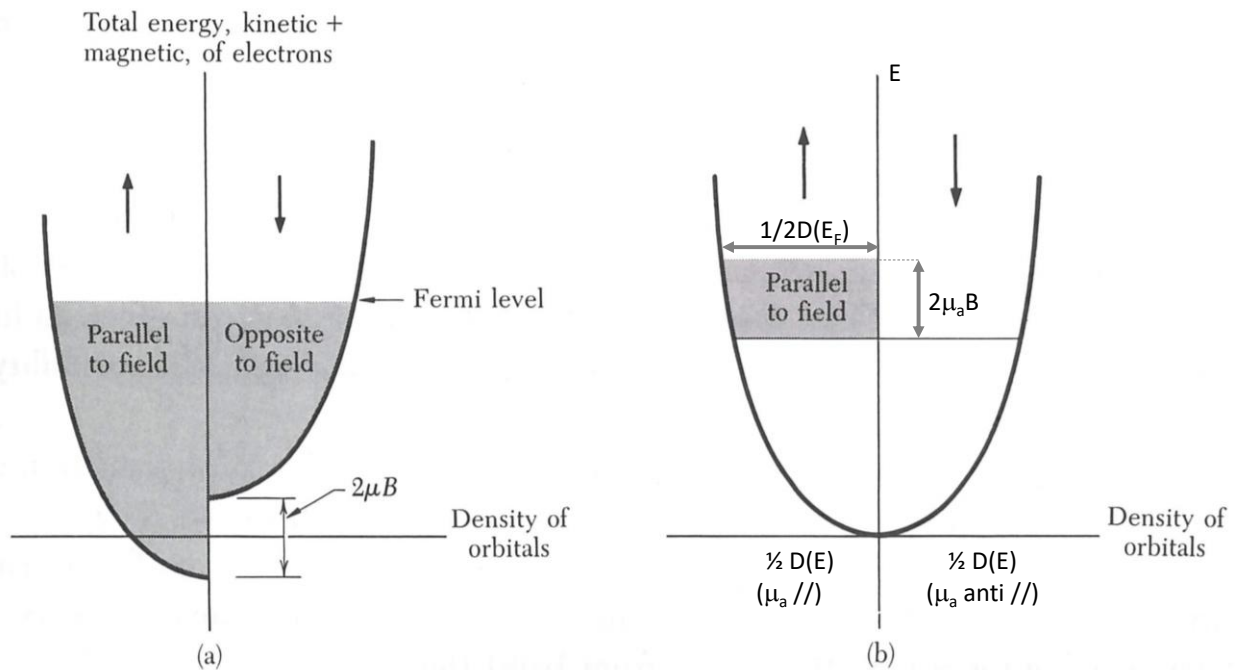
The density of states at the Fermi energy for a gas of free electrons is

$$D(E_F) = \frac{3N}{2E_F} = \frac{3N}{2k_B T_F}$$

from which

$$M_{para} = \frac{3N\mu_a^2}{2k_B T_F} B$$

known as Pauli spin magnetization.



To the magnetization due to Pauli paramagnetism, in metals must also be added the magnetization due to the diamagnetic response of the conduction electrons, which, for a gas of free electrons, Landau has shown to be equal to $-1/3$ of the paramagnetic contribution:

$$M_{dia} = -\frac{1}{3} \frac{3N\mu_a^2}{2k_B T_F} B$$

for which it turns out

$$M = M_{dia} + M_{para} = \frac{N\mu_a^2}{k_B T_F} B$$

There are also additional contributions due to the

- diamagnetism of atomic cores
- the structure of the conduction band
- some effects related to electron-electron interaction.

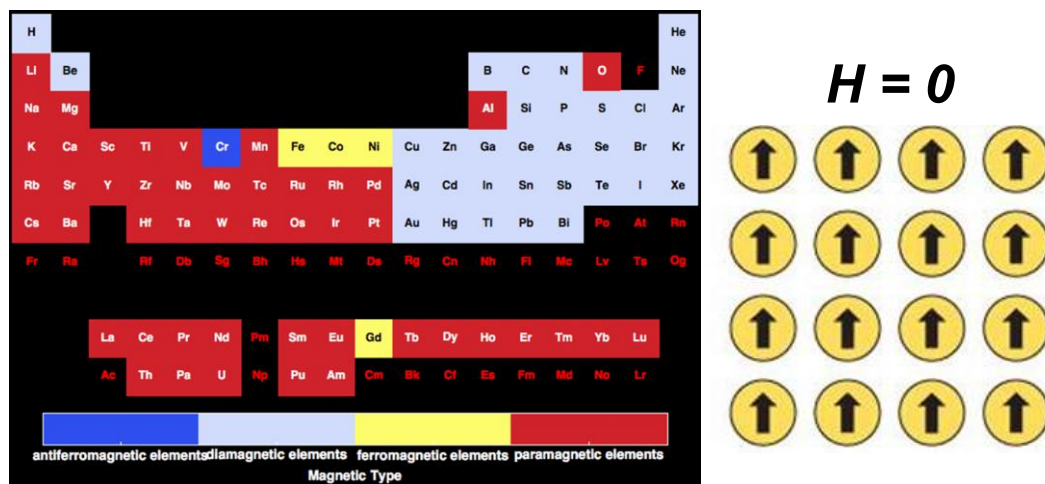
These effects can increase the resulting magnetization by up to 75% (case of Na).

Ferromagnetism.^[Kittel]

Atoms of ferromagnetic materials, such as paramagnetic materials, possess a permanent magnetic moment.

In contrast to paramagnetic materials, in which spin moments or net orbital moments are aligned in random directions in the absence of an external magnetic field, in ferromagnetic materials the presence of unbalanced spin moments leads to the formation of permanent magnetization even in the absence of an external magnetic field .

Typical examples of these materials are Fe in its phase α (BCC), Co, Ni, and Gd.



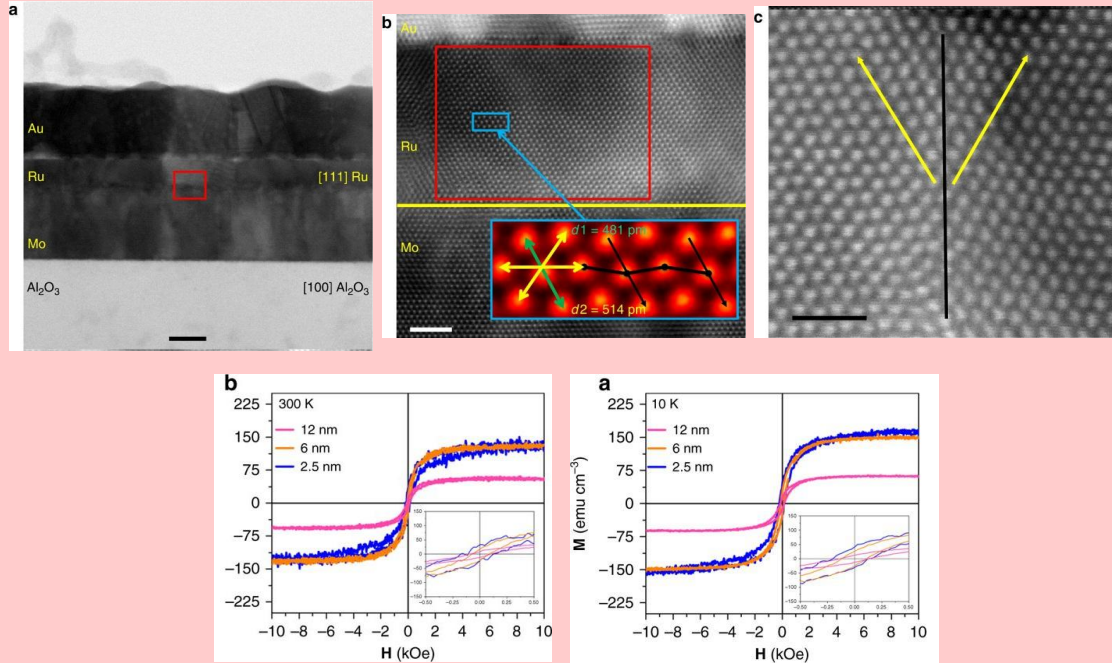
The table above refers to the magnetic properties of materials in the form of a single element and in their thermodynamically stable phase. In particular, only three elements are ferromagnetic at room temperature (Fe, Co and Ni), while Gd has a Curie temperature of 293 K. However, the formation of new compounds (such as metal alloys or the formation of oxides) in some cases leads to magnetic properties of a different type.

It is interesting the case of Ru, paramagnetic in the HCP phase, but which can become ferromagnetic if forced into a metastable tetragonal phase with a body centered, as in the case shown in the figure, where a nanometric film of Ru has grown epitaxially¹⁵, in order to induce the strain necessary to stabilize the tetragonal phase (as indicated by the XRD and TEM / STEM measurements).

This film has a saturation magnetization of 148 emu/cm^3 at room temperature and 160 emu/cm^3 at 10 K. By increasing the thickness of the film, there is relaxation of the strain and a consequent dilution of the magnetization. The figures below show the characterization of the tetragonal Ru structure by STEM analysis of a sample dissected along the direction [001] of the Al_2O_3 substrate:

¹⁵ Nature Communications, volume 9, Article number: 2058 (2018).

(a) STEM image annular bright field (ABF) of the multilayer. The bar scale is 10 nm. (b) High-angle annular dark field (HAADF) STEM at the Mo–Ru interface, with the inset showing the HAADF STEM for the direction [111] of the Ru. The distortion of the lattice is evidenced by the dashed black line. The bar scale is 2 nm. (c) HAADF STEM zoom of a Ru grain edge. The bar scale is 1 nm.



The maximum possible magnetization for these materials is saturation magnetization (M_s), usually reported per unit volume (in A/m):

$$M_s = \mu_a N \quad [M_s] = [\mu_a][N] = Am^2 \frac{1}{m^3}$$

or per unit mass of the material or magnetic centres only (in $A \, m^2/kg$)¹⁶:

$$M_s' = \mu_a N \frac{1}{\rho} \quad [M_s'] = [\mu_a][N] \left[\frac{1}{\rho} \right] = Am^2 \frac{1}{m^3} \frac{m^3}{kg}$$

It corresponds to a magnetic field of saturation B_s present inside the material even if the external magnetizing field is zero.

Some typical magnetization values for single magnetic center in metallic Fe, Co and Ni are:

μ_a	Fe	Co	Ni
μ_B	2.22	1.72	0.6

The exchange interaction.^[Ziman]

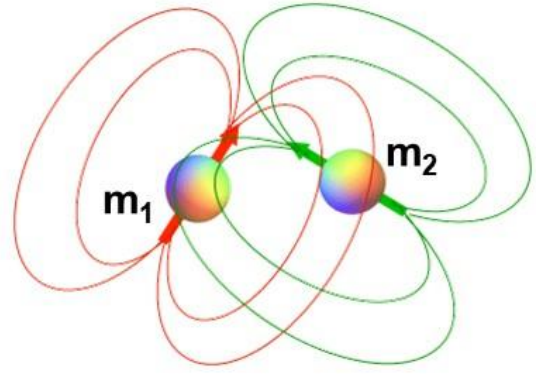
At the level of classical electromagnetism, the interaction energy between two magnetic dipoles m_1 and m_2 placed at a distance r is

¹⁶ In a generic magnetic material, composed of several elements, the magnetic centers are only the elements responsible for the magnetic response. For example, in a FeZn alloy, the magnetic centers are the atoms of Fe.

$$U_D \propto \frac{\bar{m}_1 \cdot \bar{m}_2}{r^3}$$

However, classical interaction energy is not sufficient to account for ferromagnetism, i.e. the fact that in ferromagnetic solids the magnetic moments of individual atoms spontaneously align with each other.

For example, if we assume a typical magnitude for the atomic magnetic moment $m \sim 10^{-29} \text{ Wb m}$ (the Weber - symbol Wb - is the S.I. unit of the magnetic flux, with $1 \text{ Wb/m}^2 = 1 \text{ T}$) and $r \sim 1 \text{ Å}$, we get $U_D \sim 10^{-23} \text{ J}$ which corresponds to the thermal quantum at a temperature of $\sim 1.3 \text{ K}$: above this temperature the thermal agitation would prevent any magnetic ordering.



Heisenberg and Dirac have shown that ferromagnetism, or the tendency of atomic magnetic moments to order spontaneously even in the absence of a magnetizing field, is the consequence of a quantum mechanical process attributable to the minimization of Coulomb interactions between the electric charges to which the spins responsible for the magnetic moment are associated.

Through this quantum theory, it is possible to arrive at an exchange energy between magnetic moments that is comparable with the Coulomb energy of interaction between two electric charges placed at the same distance:

$$U_C = \frac{e^2}{4\pi\epsilon_0 r^2}$$

If one assumes $r \sim 1 \text{ Å}$ as done above, gets $U_C \approx 10^{-18} \text{ J}$, which corresponds to the thermal quantum at a temperature of 10^5 K .

Qualitatively, the theory is explained by the advantage that one has from the point of view of the Coulomb repulsion to have electrons with spins aligned instead of paired: in the first case, in fact, the electrostatic repulsion is lower.

Heisenberg model^[Ziman]

According to Heisenberg's model, the Hamiltonian of interaction (or Hamiltonian of exchange energy) between the S_i and S_j spins of two atoms (to which the angular momenta $\hbar\bar{S}_i$ and $\hbar\bar{S}_j$) correspond can be written as:

$$H_{ex} = -2J\bar{S}_i \cdot \bar{S}_j$$

where J is the exchange integral, a quantity related to the overlap of the charge distribution of atoms i and j , and usually negligible when the interaction occurs at the level of thirds or sometimes even close primes.

If we consider the simple case of a system with two spins (such as a hydrogen atom with two electrons), the eigenfunction of the system will have a different structure depending on whether the spins are oriented in an antiparallel (singlet) or parallel (triplet) way. In fact, by Pauli's principle, for electrons (fermions) an antisymmetric spin function (singlet) must correspond to a symmetric orbital wave function of the type

$$\Psi_s(r_1, r_2) = \frac{1}{\sqrt{2}} [\phi_1(r_1)\phi_2(r_2) + \phi_2(r_1)\phi_1(r_2)] [\chi_\alpha(1)\chi_\beta(2) - \chi_\alpha(2)\chi_\beta(1)]$$

while a symmetric spin function (triplet) corresponds to an antisymmetric orbital wave function, of the type

$$\Psi_{as}(r_1, r_2) = \frac{1}{\sqrt{2}} [\phi_1(r_1)\phi_2(r_2) - \phi_2(r_1)\phi_1(r_2)] \begin{bmatrix} \chi_\alpha(1)\chi_\alpha(2) \\ \chi_\alpha(1)\chi_\beta(2) + \chi_\alpha(2)\chi_\beta(1) \\ \chi_\beta(1)\chi_\beta(2) \end{bmatrix}$$

Where ϕ_i are the single-electron orbital wave functions, and χ_α and χ_β the spin functions "up" and "down" for electrons 1 and 2 with coordinates r_1 and r_2 .

A wave function is antisymmetric if the exchange of coordinates r_1 of 1 with coordinates r_2 of 2 leads to a sign change of the function itself, and if $r_1 = r_2$ is posed, the wave function is zero. This means that there is no probability of finding the two electrons with the same spin in the same position at the same time (which represents the statement of Pauli's principle), while this does not apply to the two electrons with antiparallel spin.

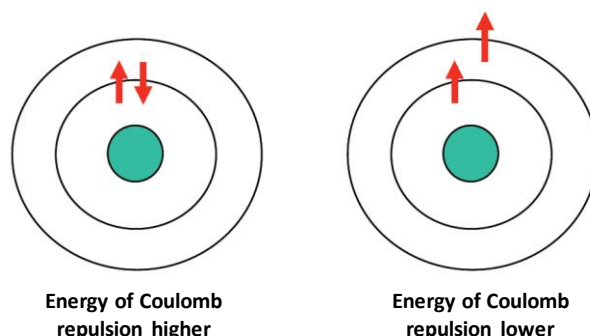
It follows that the electrostatic energy of the system depends on the orientation of the spins, and the difference in energy between the two different orientations is precisely the exchange energy.

One can write the energy associated with the two electrons in the parallel or antiparallel spin configuration as

$$E = I_1 + I_2 + K_{12} \pm J_{12}$$

Where you have $+J_{12}$ for the singlet and $-J_{12}$ for the triplet. Therefore, if J_{12} is positive, the lower energy level is for the triplet configuration:

$$E = I_1 + I_2 + K_{12} - J_{12}$$



The shape of J_{12} is such that the parallel alignment of the spins lowers the energy of:

$$J_{12} = \frac{e^2}{4\pi\epsilon_0} \int \phi_1^*(r_1)\phi_2^*(r_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \phi_2(r_1)\phi_1(r_2) d^3r_1 d^3r_2$$

where there is a $\sim 1/r$ dependence which, integrated, leads to the $\sim 1/R_2$ dependence mentioned above about the classical Coulomb repulsion energy.

The Weiss model.

In 1907, Weiss developed a theory in which the presence of an effective field within ferromagnetic materials is assumed, such as to generate spontaneous spin ordering.

On the atomic scale, the magnetic moments of ferromagnets are assumed to be due to spin angular momentum alone, due to the quenching of orbital angular momentum from the crystalline field. In fact, Fe-period elements or Gd in its metallic form exhibit ferromagnetism, thus external shells electrons are involved, which are subjected to the crystalline field effects.

Assume therefore to have a paramagnet with an N concentration of spins S : if the spins tend to align, then in ferromagnetic materials we can postulate the presence of an interaction between spin, which can be treated as a real magnetic field called B_E exchange field (or Weiss field or molecular field). Since thermal agitation is opposed to spin ordering, the exchange field must be quite intense, with values close to $10^3 T$, or about 10^4 times higher than the fields associated with naturally occurring magnetic moments.

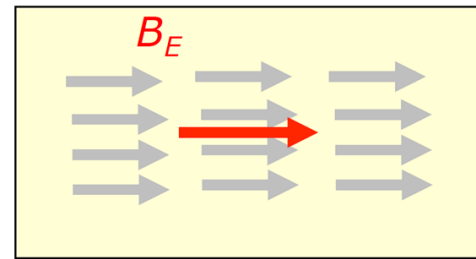
The "classical" energy of atomic magnetic moments $\mu_a = p\mu_B$ will be affected by an overall field

$$U_m = -\bar{\mu}_a \cdot (\bar{B}_a + \bar{B}_E)$$

and the M_T torque felt by the dipole (favorable to its orientation in the direction of the field) will be

$$\bar{M}_T = \bar{\mu}_a \times (\bar{B}_a + \bar{B}_E)$$

where B_a is the magnetic field in the material net of B_E , which is not a true magnetic field that obeys Maxwell's equations.



The resulting magnetization in the material, on the dimensional scale where all magnetic moments are aligned,¹⁷ will clearly depend on the intensity of B_E , and in Weiss' theory it is assumed that the two quantities are directly proportional:

$$B_E = \lambda M$$

where λ is called the molecular field coefficient or Weiss coefficient, and is independent of T .

¹⁷ As will be seen below, this happens within the individual magnetic domains.

At the phenomenological level, ferromagnetism can be explained very well by Weiss theory, although the origin of a high Weiss field is not accessible by the theory itself.

In the paramagnetic phase, the magnetization can be written as¹⁸

$$M = \chi_p (B_a + B_E) \quad (\text{c.g.s.})$$

$$\mu_0 M = \chi_p (B_a + B_E) \quad (\text{S.I.})$$

that is, the applied field will induce a magnetization which in turn will induce a Weiss field that will help induce a certain magnetization.

At the microscopic level, the self-consistent relationship between the Weiss field and magnetization is:¹⁹

$$M = N \langle \mu_a \rangle = N \mu_a \frac{\int_0^\pi \exp\left(\frac{\mu_a (B_a + \lambda M)}{k_B T}\right) \cos \theta \sin \theta d\theta}{\int_0^\pi \exp\left(\frac{\mu_a (B_a + \lambda M)}{k_B T}\right) \sin \theta d\theta}$$

If one assumes that the material in the paramagnetic phase can be described by Curie's law, then the susceptibility is

$$\chi_p = \frac{C}{T}$$

Substituting in M yields

$$\left. \begin{array}{l} \chi_p = \frac{C}{T} \\ B_E = \lambda M \end{array} \right\} \rightarrow M = \frac{C}{T} (B_a + \lambda M) \rightarrow \frac{M}{B_a} = \frac{C}{T} \frac{T}{T - \lambda C} \rightarrow$$

$$\chi = \frac{M}{B_a} = \frac{C}{(T - C\lambda)} \quad (\text{c.g.s.})$$

From this relation it is observed that the susceptibility diverges to $T_C = C\lambda$, so one can write the Curie-Weiss law in the following form

$$\chi = \frac{C}{T - T_c}$$

where

$$T_c = C\lambda$$

from which

$$\lambda = \frac{T_c}{C} = \frac{k_B T_c}{N \mu_B^2} \quad \text{per } S = \frac{1}{2}$$

¹⁸ It is also assumed in this case that $B = \mu_0 H + \mu_0 M \approx \mu_0 H$ since $\chi_m \ll 1 \rightarrow \mu_r \sim 1$ and M is very small.

¹⁹ See the derivation of the Debye equation in P07.

and for a *generic* S you get

$$\lambda = \frac{T_c}{C} = \frac{3k_B T_c}{Ng^2 S(S+1)\mu_B^2}$$

Below T_c there is ferromagnetic behavior.

For Fe, $T_c = 1063 \text{ K}$ (experimental), $\mu_a = 2.2 \mu_B$ (experimental), and $N = 8.54 \cdot 10^{28} \text{ m}^{-3}$ from which is $\lambda \sim 5000$ and a Weiss field of 10^3 T .

The exchange integral J is directly related to T_c , since it determines the intensity of the interaction between atomic magnetic moments and thus establishes the threshold beyond which thermal agitation prevails over the spontaneous tendency to order. Assuming that the magnetic center interacts only with its first neighbor z in a crystal, it is possible to write

$$J \approx \frac{3k_B T_c}{2zS(S+1)}$$

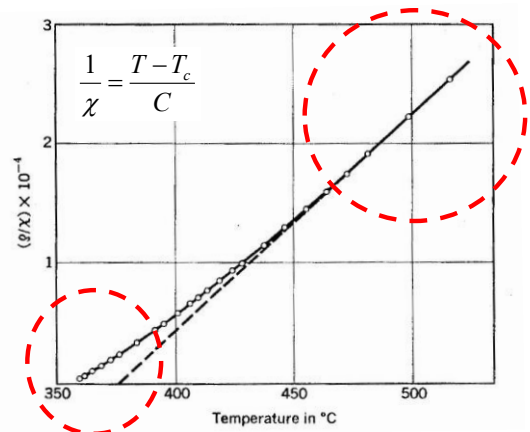
The susceptibility for temperatures above the Curie temperature.

The figure on the side shows the reciprocal of susceptibility per gram of Ni (ρ is the density)²⁰ near the Curie temperature of the metal (358 °C), measured experimentally (white circles). The dashed line is a linear extrapolation of the high-temperature section, where it is seen that the intercept corresponds to a higher value of T_c .

As a first approximation, for $T \gg T_c$ the Curie-Weiss law can be corrected in the form

$$\chi = \frac{C}{T - \theta}$$

where $\theta > T_c$.



At temperatures close to T_c , however, the Curie-Weiss law satisfactorily describes the behavior of ferromagnetic materials, but the detailed calculation provides a dependency such as:

$$\chi \propto \frac{C}{(T - T_c)^\gamma}$$

With a "critical" exponent $\gamma \sim 1.2 - 1.6$,²¹ as indicated in the table:

²⁰ Being $\chi_m = M / H$, if we use magnetization per unit ("mass magnetization") $[M] = \text{Am}^2/\text{kg}$ and $[H] = \text{A/m}$, derive $[\chi] = \text{m}^3/\text{kg}$, which is the "mass susceptibility" (see table at the beginning of the chapter).

²¹ See the meaning of the critical exponent in the 2nd order phase transitions described by the Ginzburg-Landau theory in P07.

Metal	γ	T_c (K)
Fe	1.33±0.015	1043
Co	1.21±0.04	1388
Ni	1.35±0.02	627.2
Gd	1.3±0.1	292.5
CrO ₂	1.63±0.02	386.5
CrBr ₃	1.215±0.02	32.56

Magnetization at temperatures below Curie temperature.

The Weiss model also allows to derive the dependence of spontaneous magnetization in a ferromagnet at varying temperature between 0 K and T_C .

Curie's law was obtained by assuming that the argument of the hyperbolic cotangent in Brillouin's expression was $x \ll 1$, and truncating the power series expansion of the *coth* at the second-order term.

Using instead, the complete Brillouin expression for the magnetization of atoms with $S = 1/2$, and replacing B with only the exchange field $B_E = \lambda M$ (i.e., omitting the applied magnetic field, so as to consider the spontaneous ordering of magnetic moments), we have

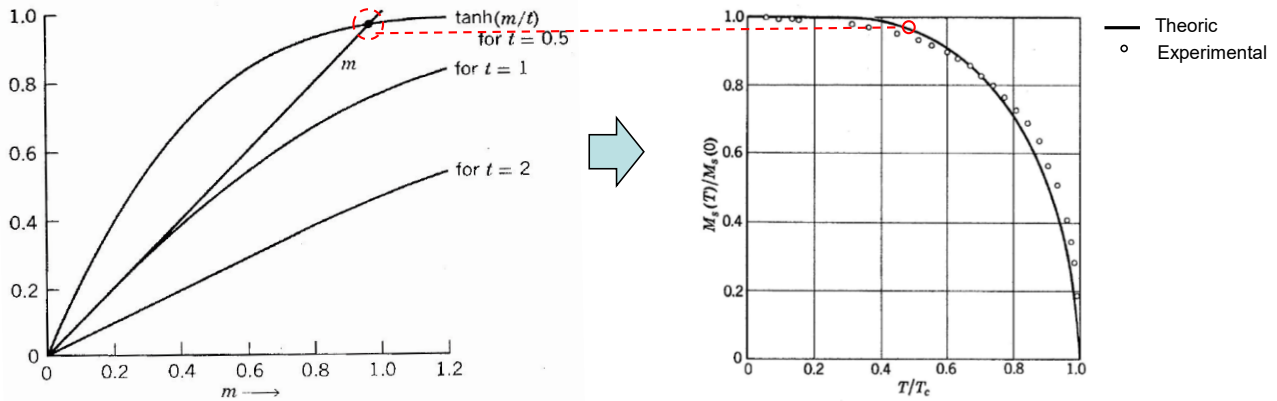
$$\left. \begin{aligned} M &= N\mu_a \tanh(x) \\ x &= \frac{\mu_a B}{k_B T} \\ B &= B_E = \lambda M \end{aligned} \right\} M = N\mu_a \tanh\left(\frac{\mu_a \lambda M}{k_B T}\right)$$

Using small sizes as follows:

$$\left. \begin{aligned} m &= \frac{M}{N\mu_a} \\ t = \frac{T}{T_c} &= \frac{k_B T}{N\mu_a^2 \lambda} \end{aligned} \right\} \rightarrow m = \tanh\left(\frac{m}{t}\right)$$

we arrive at an expression for reduced magnetization that can be solved graphically by reporting m , which is independent of T , and $\tanh(m/t)$, which obviously varies with reduced temperature. For $t < 1$, i.e. $T < T_c$, solutions are possible, from which the theoretical value of M is derived.

In this way we derive the theoretical curve of M vs T shown in the figure on the side (solid line) for the Ni ($S = 1/2$). The experimental data (white circles) are discreetly reproduced from the experimental trend.



It is clearly observed that $M \rightarrow 0$ for $T_c \rightarrow T$ as expected for a continuous phase transition of the 2nd order in which the order parameter is M . [Strobl]

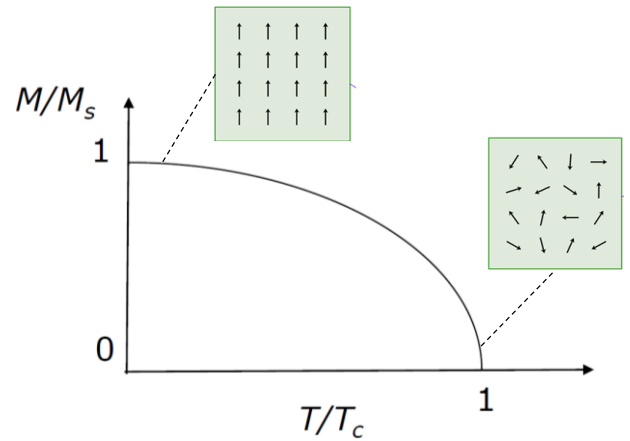
As the temperature decreases, there will be an increasing number of magnetic moments aligned with each other, according to

$$M = N \langle \mu_a \rangle = N \mu_a \frac{\int_0^\pi \exp\left(\frac{\mu_a \lambda M}{k_B T}\right) \cos \theta \sin \theta d\theta}{\int_0^\pi \exp\left(\frac{\mu_a \lambda M}{k_B T}\right) \sin \theta d\theta}$$

The above expression can be written in a more compact way if we introduce the Langevin function L , in analogy to what we will see in P07 for the Debye equation (orientation of polar molecules in an external electric field), for which we arrive at

$$M = N \mu_a L\left(\frac{\mu_a \lambda M}{k_B T}\right) = N \mu_a L(\alpha)$$

$$\alpha = \frac{\mu_a \lambda M}{k_B T}$$



At 0 K, the saturation magnetization coincides with the density of atomic magnetic moments for the magnetic moment itself:

$$M(T=0) = N \mu_a = N n_B \mu_B$$

with the magnetic moment $n_B \mu_B$ which should correspond to an integer number of Bohr magnetons, since $g \sim 2$ for an electron, and dependent only on the spin angular momentum which is responsible for the atomic magnetic moment.²²

²² Since $\bar{\mu}_a = -g \frac{\mu_B}{\hbar} \bar{J}$, if $\bar{J} = \bar{S}$ then along the axis of application of $\mathbf{B}$ we have $\mu_a = -g \frac{\mu_B}{\hbar} \hbar m_s = -g \mu_B m_s$

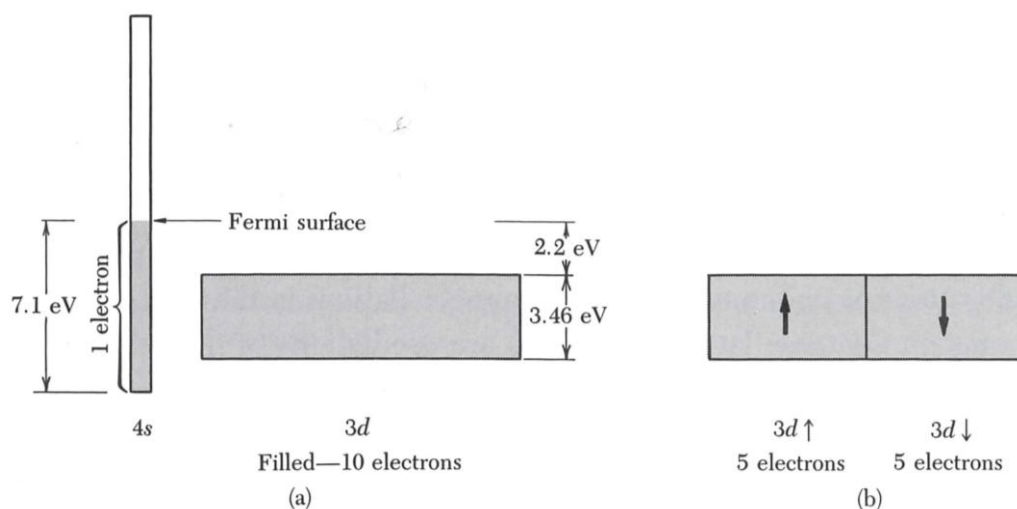
However, experimental values show non-integer numbers.

Substance	Magnetization M_s , in gauss		$n_B(0\text{ K})$, per formula unit	Curie temperature, in K
	Room temperature	0 K		
Fe	1707	1740	2.22	1043
Co	1400	1446	1.72	1388
Ni	485	510	0.606	627
Gd	—	2060	7.63	292
Dy	—	2920	10.2	88
MnAs	670	870	3.4	318
MnBi	620	680	3.52	630
MnSb	710	—	3.5	587
CrO ₂	515	—	2.03	386
MnOFe ₂ O ₃	410	—	5.0	573
FeOFe ₂ O ₃	480	—	4.1	858
NiOFe ₂ O ₃	270	—	2.4	858
CuOFe ₂ O ₃	135	—	1.3	728
MgOFe ₂ O ₃	110	—	1.1	713
EuO	—	1920	6.8	69
Y ₃ Fe ₅ O ₁₂	130	200	5.0	560

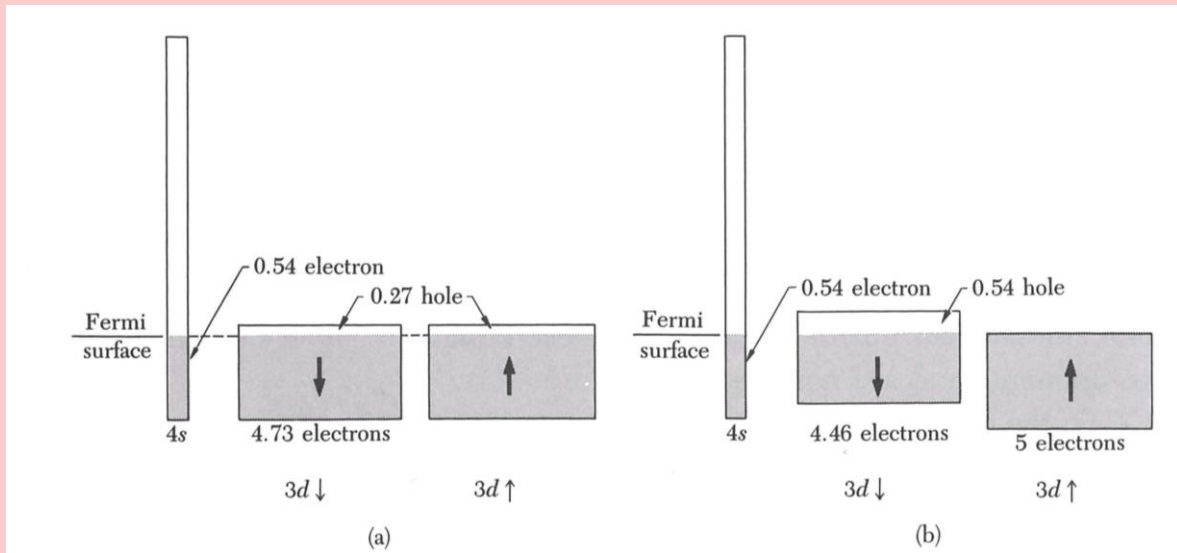
This is due to several causes:

- The spin orbit coupling carries a small contribution of orbital angular momentum (sometimes positive and sometimes negative), even where there is quenching for the crystal field.
- The interaction between atomic magnetic moments and conduction electrons (in conducting ferromagnetic materials), which induces a small magnetic moment in the latter.
- The distribution of electrons in the band structure can lead to a non-integer average number of electrons per atom in a ferromagnetic material.

The comparison of Cu (diamagnetic as it does not have d orbitals with unpaired electrons) and Ni (ferromagnetic) allows to clarify this phenomenon. The figure above shows the diagram of the 4s and 3d band structure in copper. Since the Fermi energy is above the 3d band, it contains 10 electrons per atom and is full, while the 4s contains 1 electron per atom, and is half-full. The 3d band can be divided into two subbands corresponding to electrons with opposite spin, each with 5 electrons, so the contribution to net magnetization is 0.



The figure shows the diagram of the 4s and 3d band structure of Ni above T_c (a) and 0 K (b). The Fermi energy is lower than the 3d band, so above the T_c , the 4s band contains 0.54 electrons/atom, and the 3d band contains 4.73/atom. Since above T_c the Ni is paramagnetic, in the 3d band there is the same number of electrons with spin up and down. At 0 K, on the other hand, the exchange interaction between the electrons prevails over the thermal motions, so one of the two sub-bands will be energetically favored in having the maximum number of iso-aligned spins, compatibly with the number of available orbitals, which coincides with 1/2 of the total or with 5 electrons per atom. In the other band will remain $4.73 - (5 - 4.73) = 4.46$ electrons. The result is a net magnetic moment of $5 - 4.46 = 0.54$ electrons, or $0.54 \mu_B$ (close to what is indicated in the table above). This value coincides with the gaps in the 3d band, so we often refer to the magnetic moment associated with the presence of the gaps.

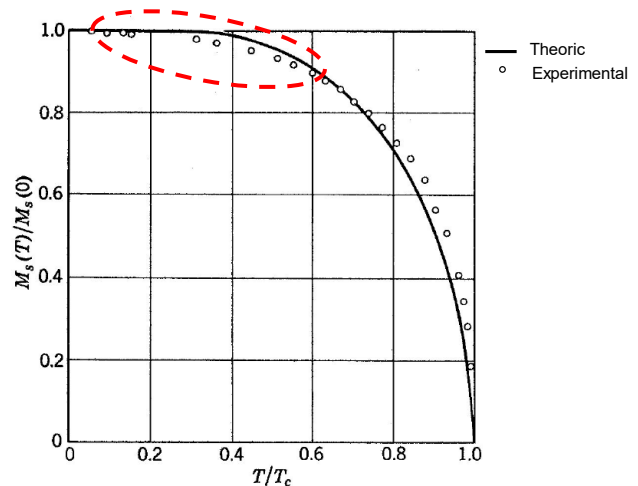


Magnoni.

Weiss's mean field theory does not reproduce well the pattern of magnetization at low temperatures, as the experimental trend shows a faster decrease of M with T in that region, of the type

$$\frac{\Delta M}{M(T=0K)} = AT^{3/2}$$

Known as Bloch's law of $T_{3/2}$ where A is a constant that depends on the material. The



explanation for this dependence lies in the presence of spin waves that lead to a more rapid misalignment of low-T spins, called magnons.

Magnons are spin waves with quantized energy. A spin wave is an oscillation of the spin direction (parallel to each other) of a ferromagnet, which propagates undulating in the material, to which is associated a characteristic energy for the system under examination or a multiple of it. Magnon physics has many similarities with that of phonons, with the difference that they are not oscillations in the relative position of crystal atoms but the relative orientation of their spins.

In the ground state (and at 0 K), spins in a ferromagnetic domain are all parallel. If one considers N spin S , then with spin angular momentum $\hbar\bar{S}$ arranged along a line, and assumes that the coupling constant J is different from 0 only for the first neighbors, then one can write the overall Heisenberg coupling energy as

$$U = -2J \sum_{p=1}^N \bar{S}_p \cdot \bar{S}_{p+1}$$

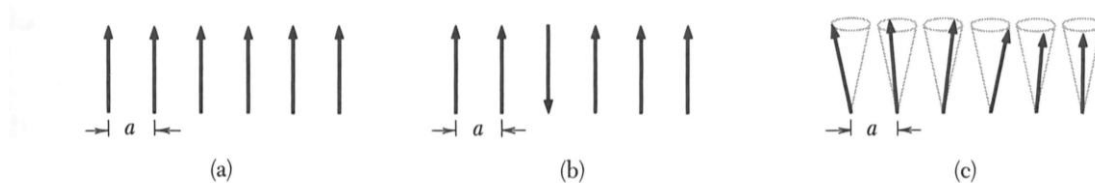
Being all parallel, at 0 K (ground state of the magnone), treating for simplicity S as a classical vector, we have

$$U_0 = -2JNS^2$$

An excitation of the system could correspond to the reversal of a spin in the row (see figure), so for two spin pairs the exchange interaction would change sign and there would be an energy

$$U_1 = -2JNS^2 - 2(-2JS^2) + 2(2JS^2) = U_0 + 8JS^2$$

It is not said that this energy cost is sustainable at low temperatures, where the experimental trend of $T^{3/2}$ is observed. On the other hand, an excitation with a lower energy cost can be obtained if the "overturning" of a spin is distributed over all the spins in the row by a precession motion:



To find the energy associated with this precession motion, first consider the p -th S_p spin, to which a magnetic moment will be associated.

$$\bar{\mu}_p = -g\mu_B\bar{S}_p$$

which allows the contribution of the p -th term to be rewritten to U as

$$\begin{aligned}
-2\bar{J}\bar{S}_p \cdot (\bar{S}_{p-1} + \bar{S}_{p+1}) &= 2\frac{J}{g\mu_B} \bar{\mu}_p \cdot (\bar{S}_{p-1} + \bar{S}_{p+1}) = \\
&= -\bar{\mu}_p \cdot \left(-2\frac{J}{g\mu_B} \right) (\bar{S}_{p-1} + \bar{S}_{p+1}) = -\bar{\mu}_p \cdot \bar{B}_p
\end{aligned}$$

where the last member shows how the whole can be expressed in the form of the interaction energy between the magnetic moment associated with the spin and an effective magnetic field (exchange) $\bar{B}_p$ acting on it:

$$\bar{B}_p = \left(-2\frac{J}{g\mu_B} \right) (\bar{S}_{p-1} + \bar{S}_{p+1})$$

From the laws of classical mechanics, the variation of angular momentum $\hbar\bar{S}_p$ in time is given by the torque due to $\bar{B}_p$:

$$\begin{aligned}
\hbar \frac{d\bar{S}_p}{dt} &= \bar{\mu}_p \times \bar{B}_p \\
\rightarrow \frac{d\bar{S}_p}{dt} &= \bar{\mu}_p \times \left(-2\frac{J}{\hbar g\mu_B} \right) (\bar{S}_{p-1} + \bar{S}_{p+1}) = -g\mu_B \bar{S}_p \times \left(-2\frac{J}{\hbar g\mu_B} \right) (\bar{S}_{p-1} + \bar{S}_{p+1}) = \\
&= \frac{2J}{\hbar} (\bar{S}_p \times \bar{S}_{p-1} + \bar{S}_p \times \bar{S}_{p+1})
\end{aligned}$$

So a misalignment between $\bar{S}_p$ and $\bar{B}_p$ leads to a non-zero torque that propagates along the entire chain triggering the magnonic oscillation.

Reporting everything in Cartesian components we have

$$\begin{cases} \frac{dS_p^x}{dt} = \frac{2J}{\hbar} [S_p^y (S_{p-1}^z + S_{p+1}^z) - S_p^z (S_{p-1}^y + S_{p+1}^y)] \\ \frac{dS_p^y}{dt} = \frac{2J}{\hbar} [-S_p^x (S_{p-1}^z + S_{p+1}^z) + S_p^z (S_{p-1}^x + S_{p+1}^x)] \\ \frac{dS_p^z}{dt} = \frac{2J}{\hbar} [S_p^x (S_{p-1}^y + S_{p+1}^y) - S_p^y (S_{p-1}^x + S_{p+1}^x)] \end{cases}$$

If the oscillation of $\bar{S}_p$ is small, then $S_x, S_y \ll S^{23}$ and one can simplify the system by assuming $S_z \sim S$ and neglecting the terms of order $S_x S_y$:

$$\begin{cases} \frac{dS_p^x}{dt} \approx \frac{2J}{\hbar} S [2S_p^y - S_{p-1}^y - S_{p+1}^y] \\ \frac{dS_p^y}{dt} \approx -\frac{2J}{\hbar} S [2S_p^x - S_{p-1}^x - S_{p+1}^x] \\ \frac{dS_p^z}{dt} \approx \frac{2J}{\hbar} [0 - 0] \end{cases}$$

²³ By convention, z is assumed as the direction of spin alignment.

This system of equations can have solutions such as

$$\begin{cases} S_p^x = u \exp[i(pka - \omega t)] \\ S_p^y = v \exp[i(pka - \omega t)] \end{cases}$$

where u and v are constants, p is the index that identifies the spin in the row, and a is the spacing between the spins.

Substituting the test functions in the system of equations gives

$$\begin{cases} -i\omega u \exp[i(pka - \omega t)] = \frac{2J}{\hbar} S[2v \exp[i(pka - \omega t)] - v \exp[i((p-1)ka - \omega t)] - v \exp[i((p+1)ka - \omega t)]] \\ -i\omega v \exp[i(pka - \omega t)] = -\frac{2J}{\hbar} S[2u \exp[i(pka - \omega t)] - u \exp[i((p-1)ka - \omega t)] - u \exp[i((p+1)ka - \omega t)]] \end{cases}$$

$$\rightarrow \begin{cases} -i\omega u \exp[i(pka)] = \frac{2J}{\hbar} S[2v \exp[i(pka)] - v \exp[i((p-1)ka)] - v \exp[i((p+1)ka)]] \\ -i\omega v \exp[i(pka)] = -\frac{2J}{\hbar} S[2u \exp[i(pka)] - u \exp[i((p-1)ka)] - u \exp[i((p+1)ka)]] \end{cases}$$

$$\rightarrow \begin{cases} -i\omega u = \frac{2J}{\hbar} S[2v - v \exp[-i(ka)] - v \exp[i(ka)]] = \frac{2J}{\hbar} S\left[2v - 2v \frac{\exp[-i(ka)] + \exp[i(ka)]}{2}\right] \\ -i\omega v = -\frac{2J}{\hbar} S[2u - u \exp[-i(ka)] - u \exp[i(ka)]] = -\frac{2J}{\hbar} S\left[2u - 2u \frac{\exp[-i(ka)] + \exp[i(ka)]}{2}\right] \end{cases}$$

$$\rightarrow \begin{cases} -i\omega u = 4 \frac{J}{\hbar} S v (1 - \cos ka) \\ -i\omega v = -4 \frac{J}{\hbar} S u (1 - \cos ka) \end{cases}$$

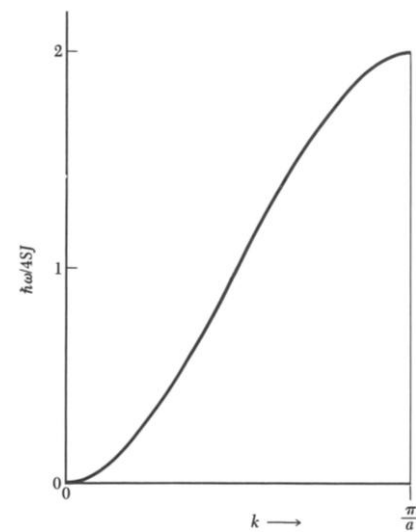
$$\rightarrow \begin{cases} i\omega u + 4 \frac{J}{\hbar} S (1 - \cos ka) v = 0 \\ i\omega v - 4 \frac{J}{\hbar} S (1 - \cos ka) u = 0 \end{cases}$$

To this system corresponds the secular determinant

$$\begin{vmatrix} i\omega & 4 \frac{J}{\hbar} S (1 - \cos ka) \\ -4 \frac{J}{\hbar} S (1 - \cos ka) & i\omega \end{vmatrix} = 0$$

$$\rightarrow -\omega^2 + \left(4 \frac{J}{\hbar} S (1 - \cos ka)\right)^2 = 0$$

$$\rightarrow \hbar\omega = 4JS(1 - \cos ka)$$



However, the dispersion relation shown in the figure on the side is valid only for the ideal case treated in this example, of a spin linear chain with exchange interaction limited to its first neighbors. Even in realistic cases of spins arranged along a three-dimensional lattice, if the exchange

interaction is limited to the first neighbors, the dispersion relation has a similar pattern to what was found in this simplified case.

Replacing $\hbar\omega$ in the system leads to the following solution

$$\begin{cases} i4JS(1-\cos ka)u + 4JS(1-\cos ka)v = 0 \\ i4JS(1-\cos ka)v - 4JS(1-\cos ka)u = 0 \end{cases} \rightarrow \begin{cases} iu + v = 0 \\ iv - u = 0 \end{cases} \rightarrow \begin{cases} v = -iu \\ v = \frac{u}{i} = -iu \end{cases}$$

and test solutions

$$\begin{cases} S_p^x = u \exp[i(pka - \omega t)] \\ S_p^y = -iu \exp[i(pka - \omega t)] \end{cases}$$

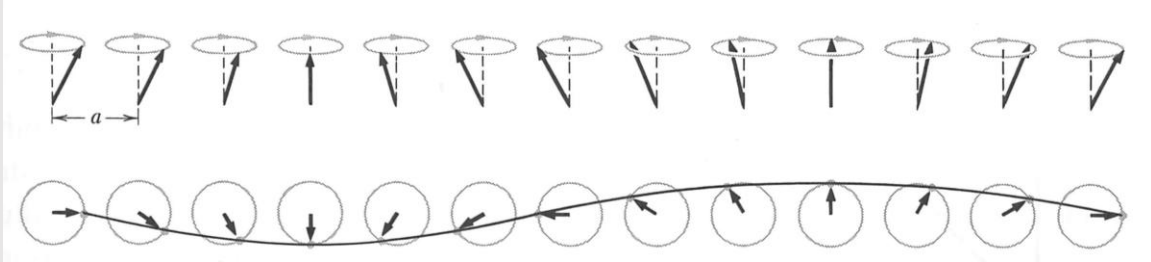
This result corresponds to a precession with circular orbit about the z-axis.

$$x = (pka - \omega t) \rightarrow \begin{cases} S_p^x = u e^{ix} = u \left(\frac{e^{ix}}{2} + \frac{e^{ix}}{2} + \frac{e^{-ix}}{2} - \frac{e^{-ix}}{2} \right) = u \cos x + u \frac{i}{i} \left(\frac{e^{ix}}{2} - \frac{e^{-ix}}{2} \right) = u \cos x + i u \sin x \\ S_p^y = -iu e^{ix} = -iu \left(\frac{e^{ix}}{2} + \frac{e^{ix}}{2} + \frac{e^{-ix}}{2} - \frac{e^{-ix}}{2} \right) = -iu \cos x - iu \frac{i}{i} \left(\frac{e^{ix}}{2} - \frac{e^{-ix}}{2} \right) = -iu \cos x + u \sin x \end{cases}$$

whose real components are

$$\begin{cases} \text{Re}[S_p^x] = u \cos(pka - \omega t) \\ \text{Re}[S_p^y] = u \sin(pka - \omega t) = u \cos\left(pka - \omega t - \frac{\pi}{2}\right) \end{cases}$$

where the phase relationship $\pi/2$ between the two components is evident.



The solution coincides with what would be obtained through the complete quantum mechanical treatment.

If the wavelength of the magnon is very large compared to the periodicity of the spins, or

$$ka = \frac{2\pi}{\lambda} a \ll 1$$

Then one can expand the cosine in truncated power series to the first term for which

$$\hbar\omega = 4JS(1 - \cos ka) \approx 4JS \left(1 - 1 + \frac{1}{2}(ka)^2 \right) = 2JSa^2 k^2$$

Magnon energy is quantized in the same way as photons and phonons.

$$E_k = \left(n_k + \frac{1}{2} \right) \hbar \omega_k$$

where n_k is the Magnone number with wave vector k .

Like photons and phonons, the number of excited magnons at a certain temperature is given by the Planck distribution, which applies to all physical problems in which the potential is harmonious:

$$\langle n_k \rangle = \frac{1}{\exp\left(\frac{\hbar \omega_k}{k_B T}\right) - 1}$$

Each $\Delta n_k = 1$ corresponds to the reversal of a spin, so the variation in magnetization with respect to magnetization at 0 K is proportional to the ratio

$$\frac{\Delta M}{M(0)} \propto \frac{\sum_k n_k}{N} \propto T^{3/2}$$

which in turn can be calculated to be proportional to $T^{3/2}$, in accordance with experiments. This relation is called Bloch's law of $T^{3/2}$.

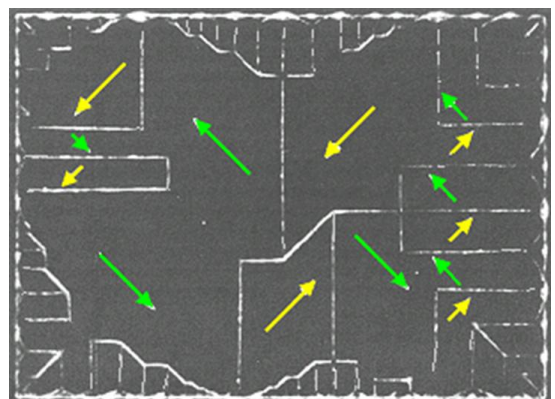
The ferromagnetic domain structure.

A "macroscopic" ferromagnetic material, also monocrystalline, can have zero initial magnetization, if cooling below the Curie temperature occurs in the absence of an external magnetic field. This depends on the spontaneous division of the material into magnetic domains, i.e. small regions within which the magnetization is different from 0, but has no direction parallel to that of magnetization in neighboring domains. These magnetic domains are called Weiss domains.

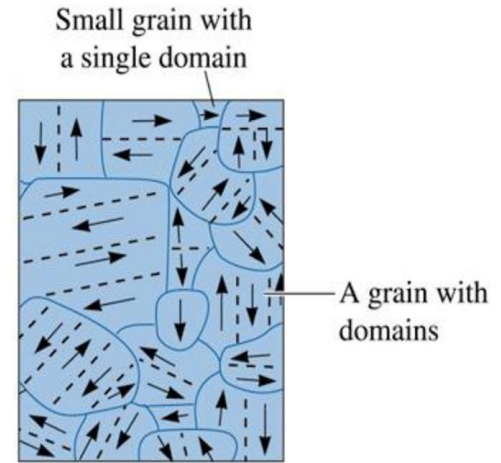
The magnetization that develops in magnetic domains in the absence of an external field is directed along some preferential directions, established by the symmetry of the crystal, and called "directions of easy magnetization". This means that, at distances much greater than atomic, the advantage in having parallel alignment of magnetic moments, due to the exchange energy, is outweighed by the tendency of the dipoles to align in an antiparallel direction with each other, as predicted by the classical model of magnetism.

The boundary between two magnetic domains, or the area in which the magnetization changes direction, is called the domain wall or Bloch wall. It consists of a gradual rotation of the direction of magnetic moments, over distances of the order of a few hundred atoms (about 300 in the case of Fe).

The domains are present both in a single crystal and in a



polycrystalline material, since there is no correlation between the crystallographic and magnetic domains, on a scale higher than that of the single crystallographic cell. The figure shows the arrangement of ferromagnetic domains in a monocrystalline nickel foil. Thus, it is evident that there is no relationship between the dimension of the Bloch domains and that of the crystalline domains in the material. The direction of magnetization in each domain, indicated by the arrow, is identified by monitoring the growth or contraction of each domain by application of an external magnetizing field. The white lines indicate the walls of the magnetic domains, and are obtained by spreading an oil containing fine iron powder on the crystal.

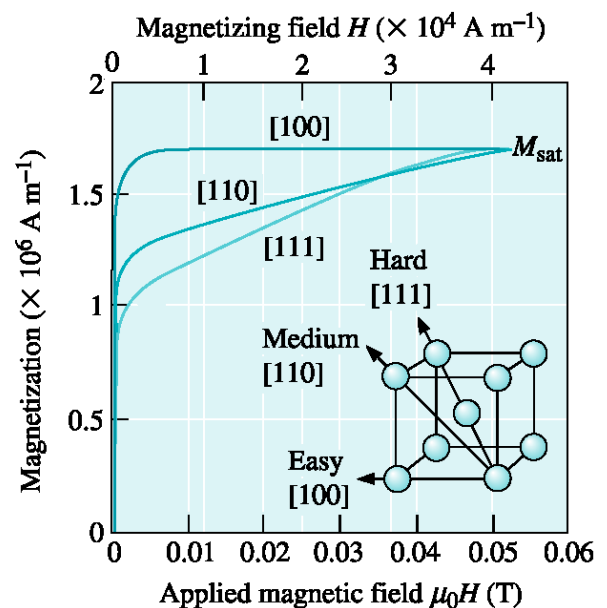


In general, magnetic domains that form spontaneously (without application of external field) and crystalline domains do not coincide, but this is true only above a critical dimension, as schematized in the figure on the side, which shows a qualitative representation of magnetic domains in a polycrystalline material (Bloch walls are indicated as dotted lines while wheat edges are indicated in blue). In addition, magnetic domains usually do not extend beyond crystallographic domains, except randomly as shown in the figure, because crystallographic directions in polycrystalline materials are random, and magnetization within each crystalline domain will always follow directions of easy magnetization.

Directions of easy magnetization and anisotropy energy.

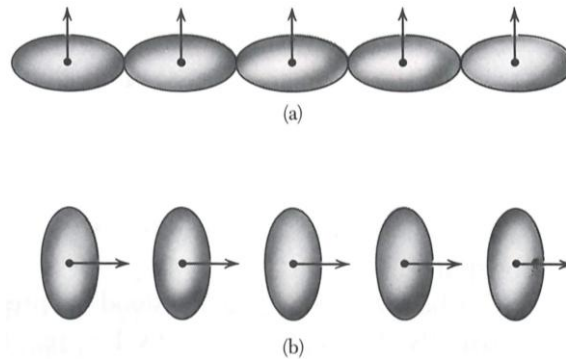
The magnetization curves for Fe are anisotropic, in fact magnetization is facilitated when the directions [100] are aligned with the field and is more difficult along the direction [111].

The difference in energy for magnetization along two different directions is called magnetic anisotropy energy. Since the exchange interaction leading to magnetic ordering is an isotropic phenomenon,²⁴ anisotropy energy originates in the crystal structure of the ferromagnet and in the spin



²⁴ Changing the direction of magnetization is equivalent to changing the orientation of the spins in the material, but this does not involve any change in the position of the atoms in the crystalline cell, so it is an isotropic process except for a secondary effect due to the asymmetry of the orbitals due to the spin-orbit interaction.

orbit coupling between S and L . The spin-orbit interaction leads to a slight asymmetry in the charge distribution, which moves with the spin in its orientation motion along the magnetizing field (see figure on the side). Since atomic distances vary depending on the crystallographic direction, some orientations will be less favored due to a higher electrostatic repulsion contribution, and vice versa, so not all magnetization directions have the same energy cost in a ferromagnetic crystal.



For example, in Co, which has H.C.P. structure, the direction of easy magnetization is that of the main axis C_6 and the anisotropy energy is expressible as a function of the angle θ between the magnetization and the main axis of symmetry:

$$U_k^{Co} = K_1' \sin^2 \theta + K_2' \sin^4 \theta$$

with K_1' and K_2' two temperature-dependent constants. When the direction of magnetization is parallel to the C_6 axis, the sine cancels out and, with it, the anisotropy energy is zero.

As shown in the figure, in the α -Fe, the directions of easy magnetization coincide with the edges of the cubic cell, and the anisotropy energy is given by

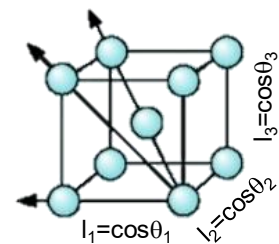
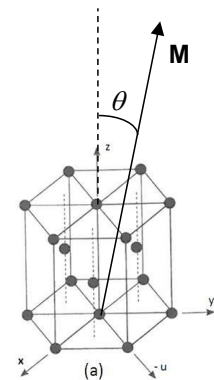
$$U_k^{\alpha-Fe} = K_1' (l_1^2 l_2^2 + l_2^2 l_3^2 + l_3^2 l_1^2) + K_2' l_1^2 l_2^2 l_3^2$$

where are the directing cosines the magnetization with respect to the three axes of the cubic cell. In this case, when the direction is parallel to one of the edges of the cell, one of the directing cosines is 1 while the other two are 0, so the anisotropy energy cancels out.

The constants K' are called anisotropy constants and are usually measured in energy per unit volume.

The transition region between domains.

The magnetic moments of the cells in the Bloch walls change direction continuously from one domain to another. The change in direction of spins at the boundary between two magnetic domains



does not occur within a single atomic plane, but can involve a layer hundreds of atomic planes thick. The reason lies in the fact that the exchange energy is lower if the spin rotation occurs gradually.

The transition region that separates two domains with different direction of magnetization in a magnetic material is called the Bloch wall.

The thickness of the Bloch wall can be obtained using a classical interpretation of the Heisenberg model,

$$H_{ex} = -2J\vec{S}_i \cdot \vec{S}_j$$

For example, if we consider two spins S between which there is an angle ϕ

$$U_{ex} = -2JS^2 \cos \phi$$

If ϕ is small, then

$$\cos \phi \approx 1 - \frac{1}{2}\phi^2 \rightarrow U_{ex} \approx -2JS^2 \left(1 - \frac{1}{2}\phi^2\right)$$

where it is seen that the presence of an angle ϕ between the two spins corresponds to an additional energy cost equal to

$$\Delta U_{ex} = JS^2 \phi^2$$

If the rotation of 180° in the direction between the spins is distributed over $N + 1$ atomic planes, the angle ϕ between two spins will be

$$\phi = \frac{\pi}{N} \rightarrow \Delta U_{ex} = JS^2 \frac{\pi^2}{N^2}$$

and the total energy for a spin row will be

$$N\Delta U_{ex} = JS^2 \frac{\pi^2}{N}$$

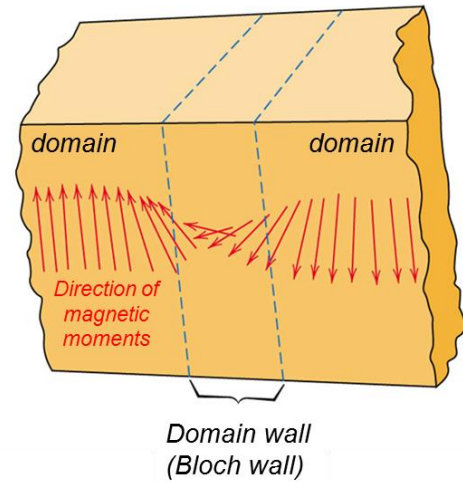
So the energy tends to 0 per $N \rightarrow \infty$, but the increase in thickness of the Bloch wall is balanced by the anisotropy energy, which is higher for spins that deviate from directions of easy magnetization.

The energy of total anisotropy U_{anis} will be proportional to the number of spins N in the row:

$$U_{anis} \propto N$$

To derive the value of N at which there is equilibrium between the gain in exchange energy and the loss in anisotropy energy, we can consider the simple case of a Bloch wall that separates two domains with magnetization in opposite directions due to spins arranged in a simple cubic crystal.

The total energy σ_{tot} associated with the formation of a Bloch wall with unit area section will be



$$\sigma_{tot} = \sigma_{ex} + \sigma_{anis}$$

where the exchange term σ_{ex} will be proportional to the number of atomic plans involved and the number of spins per unit area.

Since in a simple cubic lattice we have 1 spin for each face of the cell of side a , the spin density per unit area will be $1/a^2$, so the exchange energy density per unit area will be equal to the exchange energy for the single spin (ΔU_{ex}) for the spin density per unit area ($1/a^2$) for the number of atomic planes in the Bloch wall (N):

$$\sigma_{ex} = \frac{N \Delta U_{ex}}{a^2} = JS^2 \frac{\pi^2}{N} \frac{1}{a^2}$$

The total anisotropy energy will be proportional to the wall thickness (Na), with proportionality constant approximately equal to the "average" anisotropy constant expressed per unit volume (K):²⁵

$$\sigma_{anis} \approx NaK$$

whereby

$$\sigma_{tot} = JS^2 \frac{\pi^2}{N} \frac{1}{a^2} + KNa$$

Minimizing with respect to N is

$$\frac{d\sigma_{tot}}{dN} = -JS^2 \frac{\pi^2}{N^2} \frac{1}{a^2} + Ka = 0$$

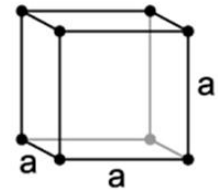
$$\rightarrow JS^2 \frac{\pi^2}{a^2} = KaN^2$$

$$\rightarrow N = \left(JS^2 \frac{\pi^2}{a^3 K} \right)^{1/2}$$

which is proportional to the root of the exchange integral J and inversely proportional to the root of the anisotropy constant K .

The energy per unit area is

$$\begin{aligned} \sigma_{tot} &= JS^2 \frac{\pi^2}{\left(JS^2 \frac{\pi^2}{a^3 K} \right)^{1/2}} \frac{1}{a^2} + K \left(JS^2 \frac{\pi^2}{a^3 K} \right)^{1/2} a = \\ &= J^{1/2} S \frac{\pi K^{1/2}}{a^{1/2}} + J^{1/2} S \frac{\pi K^{1/2}}{a^{1/2}} = \end{aligned}$$



$$\begin{aligned} \sigma_{ex} &= \frac{\text{Number of atomic planes in the Bloch wall} \times \text{Exchange energy for 1 spin} \times \text{Number of spins per unit area}}{a^2} \\ \sigma_{anis} &\approx \frac{\text{Thickness of The Bloch wall} \times \text{Average anisotropy constant (energy per unit volume)}}{a} \end{aligned}$$

²⁵ The anisotropy energy depends on the angle between the spins and the direction of easy magnetization, so in case of rotation of 180° along the row of N spin it would be necessary to perform the average on this angle. Multiplying K (expressed in energy per unit volume) by the length of part Na gives an energy σ_{anis} per wall of unit area.

$$= 2\pi \left(\frac{KJS^2}{a} \right)^{1/2}$$

The energy per unit area per single floor is

$$\begin{aligned} \frac{\sigma_{tot}}{N} &= \left(JS^2 \frac{\pi^2}{N} \frac{l}{a^2} + KNa \right) \frac{l}{N} = JS^2 \frac{\pi^2}{N^2} \frac{l}{a^2} + Ka = JS^2 \frac{\pi^2}{\left(JS^2 \frac{\pi^2}{a^3 K} \right)} \frac{l}{a^2} + Ka = \\ &= JS^2 \frac{\pi^2 a^3 K}{JS^2 \pi^2} \frac{l}{a^2} + Ka = 2aK \end{aligned}$$

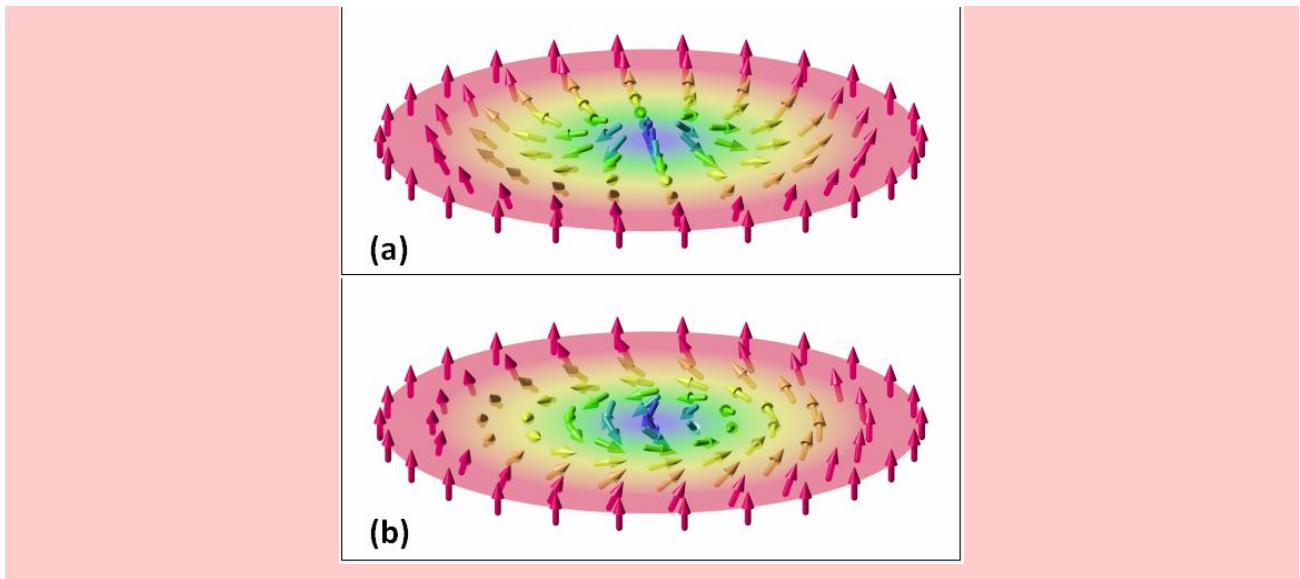
Performing the same calculation in the case of Fe (B.C.C.) for a 180° rotation of the spin in a direction perpendicular to the plane (100), we obtain a total energy equal to

$$\sigma_{tot}^{\alpha-Fe} = 2 \left(\frac{2K_1 JS^2}{a} \right)^{1/2}$$

and a number of atomic plans $N \approx 300$.

In physics, the rotation of the spin that is observed in a Bloch wall represents an example of a "soliton", or a physical quantity that can be described as a localized wave, unless there is a magnetizing field that stimulates its propagation, and in this case the encounter with another Bloch wall (another soliton) does not change its amplitude or frequency but only the phase. Solitons are observable phenomena, under appropriate conditions, in optics and mechanics and, together with skirmioni, are gaining relevance in the field of quantum technologies.

Spin in materials subject to spontaneous magnetic ordering (ferromagnets, antiferromagnets and ferrimagnets) can also be organized in local energy minimums with a "topologically" anomalous configuration, such as to be different from a soliton. These energy minimums are defined "skyrmions" and are quasiparticles corresponding to a topologically non-trivial magnetic configuration confined to a finite and localized region of the materials, generally stable or metastable with respect to its transformation into a homogeneously magnetized ground state that is equivalent to the magnetic soliton state.



The origin of magnetic domains.

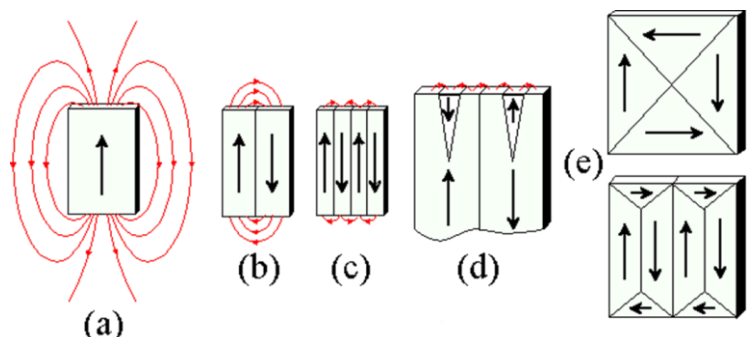
The previous calculation shows that the formation of Bloch walls has a non-zero energy cost. Therefore, if the ferromagnetic material spontaneously organizes itself into domains, there must be an energy advantage of different origin. Landau and Lishfitz have shown that the presence of magnetic domains results from the balance between the different contributions of exchange energy, anisotropy energy and magnetic energy.

In particular, the magnetic energy U_M associated with the presence of a magnetic field B in a certain volume of space V is given by:

$$U_M = \frac{1}{8\pi} \int B^2 dV$$

It is verified that, for a ferromagnet, the configuration with the lowest energy cost will tend to be the one in which the B field has the least possible spatial extension.

The figure qualitatively schematizes the field lines in a hypothetical uniaxial magnetic material²⁶ in the case of (a) single magnetic domain, (b) two and (c) four domains. It is observed that the magnetic flux density outside the material is progressively reduced as the number of nd domains increases, with a dependence that proves to be



$$U_M \approx \frac{1}{n_d}$$

²⁶ In uniaxial materials, the direction of easy magnetization coincides with the main axis of symmetry.

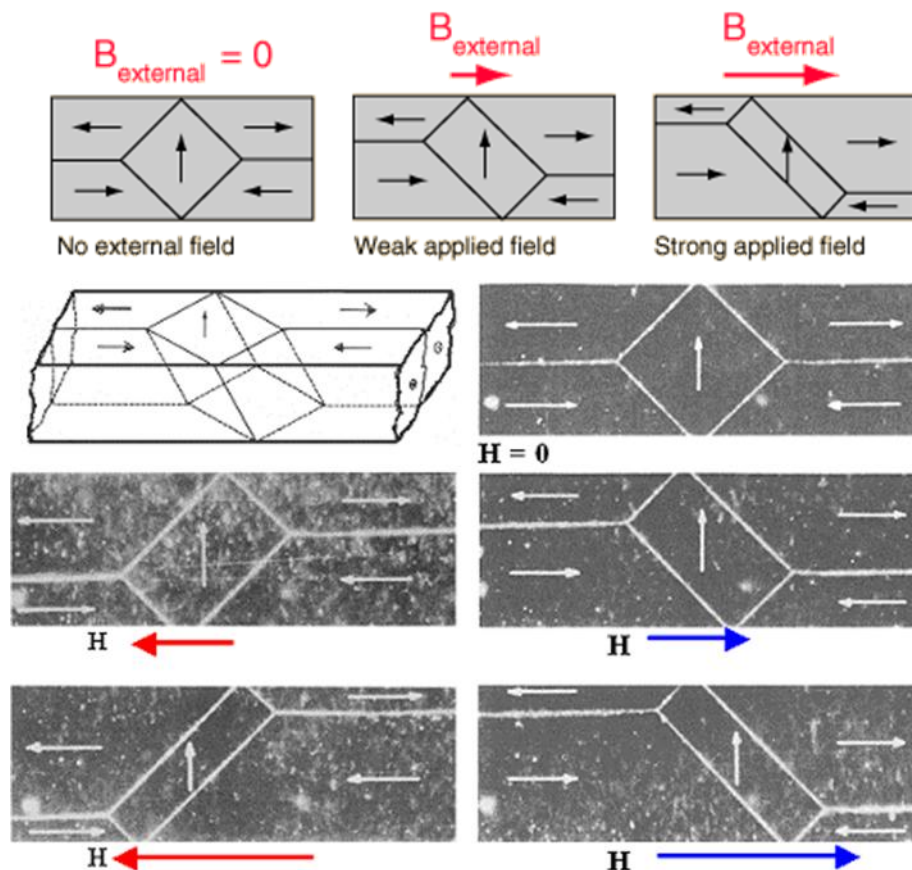
But it never cancels, while inside the material the magnetic energy remains practically unchanged. (d) In the case of uniaxial materials, inverted wedge domains may be formed to further reduce the flow density outside. (e) The domains that allow the total concatenation of the magnetic field flux, without magnetic flux permeation outside the sample, are those with wedge structure and direction of magnetization orthogonal to that of adjacent domains. These domains are called closure domains, because they complete the magnetic flux circuit entirely within the material.

Magnetization curves and hysteresis.

The increase in magnetization takes place by two independent processes:

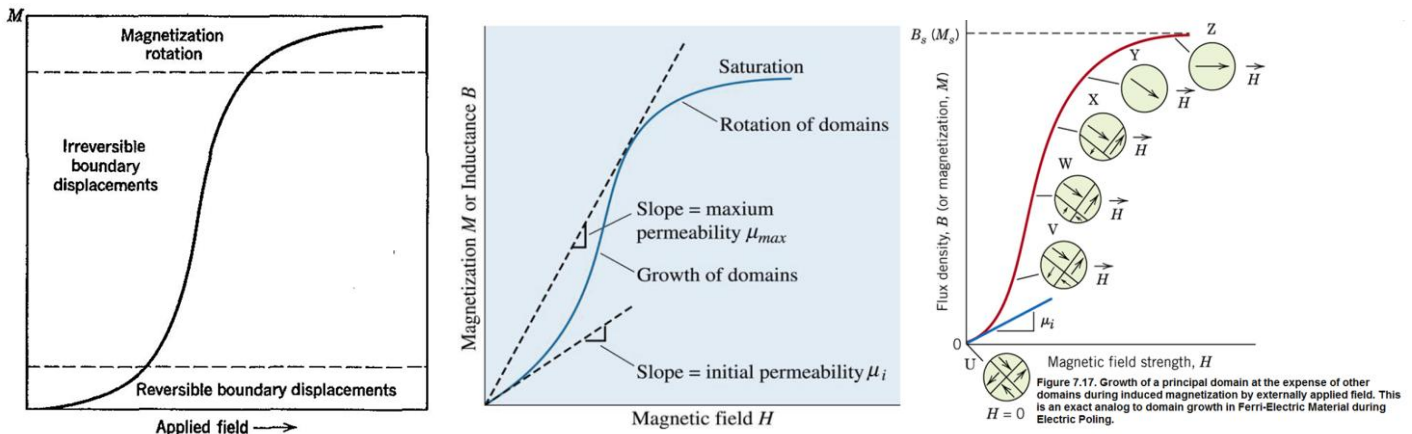
1. At moderate fields, the volume of favorably oriented domains increases at the expense of those oriented in an unfavorable way, by migration of Bloch walls.

The figures below show a diagram of the process and an experimental measurement performed on a small iron monocrystal.



2. At very strong fields, the magnetization rotates in the direction of the external field.

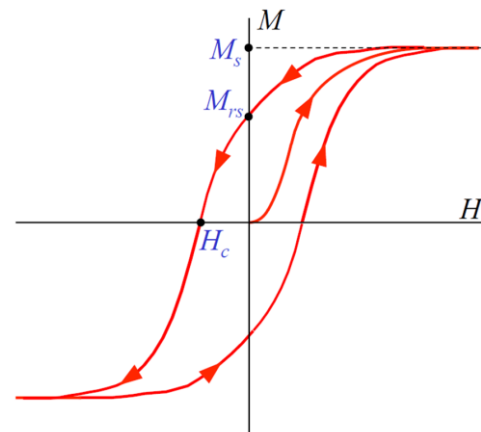
In hysteresis curves, this process begins at the inflection point of the last stroke (see figure). Another inflection point can be identified in the low-field zone, corresponding to the initial phase in which the expansion of magnetic domains is reversible.



It is therefore possible to measure a hysteresis characteristic for a certain ferromagnetic (or ferrimagnetic, see below) material, i.e. the profile traced by magnetization during a cycle of variation of the intensity of the magnetizing field applied to the material itself according to $H = 0 \rightarrow +H_{max} \rightarrow 0 \rightarrow -H_{max} \rightarrow 0$.

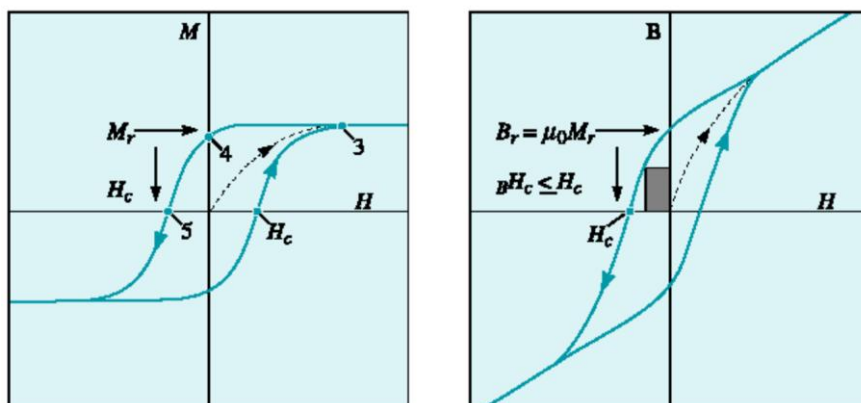
In such a cycle the following characteristic quantities can be identified:

- M_s - saturation magnetization: the maximum possible magnetization in a material, observed when all dipoles have been aligned by the magnetizing field.
- M_{rs} magnetic remanence: The magnetization that remains in a material after removal of the external field.
- Coercive field H_c : also called coercivity, the field that must be applied to the material to cancel the magnetization.



As can be seen from the figure, a ferromagnetic material can initially have a zero magnetization, or different from 0 depending on its "history" of exposure to external magnetizing fields.

Hysteresis $M-H$ corresponds to hysteresis $B-H$, which however is of less practical interest because the value of B never saturates being $B = \mu_0 H + \mu_0 M$.



Coercivity.

The coercivity can vary by as many as 7 orders of magnitude depending on the type and structure (understood as the size and dispersion of crystalline grains in other magnetic and non-magnetic dispersing phases) of the magnetic material, and its value can be modified by acting on the structure of the material.

As shown by the hysteresis cycle, magnetic permeability has a variable value depending on the strength of the applied field H , but also on the "history" of the material, since the same value of H can correspond to different values of M and different slopes. Therefore, the magnetic susceptibility for ferromagnetic materials cannot be uniquely identified, but depends on the state of spin ordering in the material. In general, one can define as the derivative of M with respect to H valued at the value of the applied field H_i :

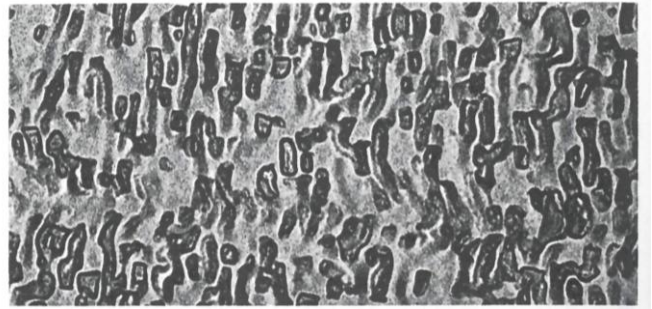
$$\chi_{ferrom} = \left. \frac{dM}{dH} \right|_{H_i} \rightarrow \mu_r = 1 + \left. \frac{dM}{dH} \right|_{H_i}$$

In materials with high coercive field, the variation of M with H will be slower, which will correspond to a low magnetic permeability, and vice versa for materials with low coercive field.

The coercive field can be decreased (and magnetic permeability increased) minimizing material defects.

Instead, a high coercive field can be achieved by structuring the material into particles the size of a single magnetic domain, but assembled so that

there are no crystalline connections between them. This can occur in a heterogeneous structure in which the magnetic particles are dispersed in a second metallic phase, as in the figure on the side referring to a section of the width of 1100 nm of the system Al(8wt%)-



Ni(14wt%)-Co(24wt%)-Cu(3wt%)-Fe(51wt%), called Alnico V. The precipitation is performed in the presence of a magnetic field, so that the nanoparticles are oriented with their major axis all in the same direction, so as to obtain an anisotropic structure.

Crystals with dimensions close to 10^{-7} - 10^{-8} m are generally single-domain, because the division into subdomains becomes energetically expensive. Therefore, in the individual domains, the rotation of the magnetization cannot occur by moving the Bloch walls, but must rotate all together, which requires more energy. In single-domain Fe particles, $H_c \approx 500$ Gauss.

Ferrimagnetism.

At the macroscopic level, the magnetic behavior of ferrimagnetic materials is similar to that of ferromagnetics.

At the microscopic level, as in ferromagnetics, the cell has a net magnetic moment, but not as large as if all magnetic centers were aligned in the same direction.

In fact, in ferrimagnetism some magnetic centers are constructively coupled, and others destructively.

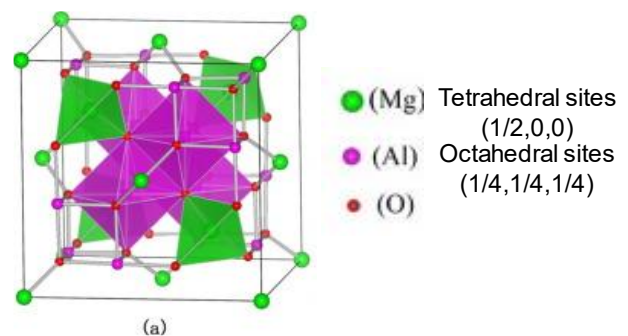
Examples of ferrimagnets are Fe_3O_4 , NiFe_2O_4 , ZnFe_2O_4 .

A common feature of ferrimagnets is that they are insulators, while many ferromagnets are conductive metals. This is useful for all applications where non-conductive magnetic materials are required, as described below.

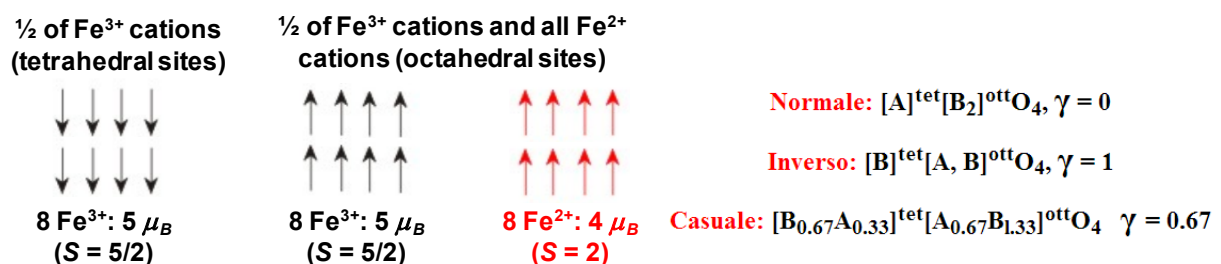
The case of ferrites.

Fe_3O_4 magnetite has a $\text{FeO}\cdot\text{Fe}_2\text{O}_3$ reverse spinel structure.

The normal structure of spinels (F.C.C. lattice) is that of MgAl_2O_4 , with divalent ions (Mg^{2+}) in tetrahedral sites and trivalent ions (Al^{3+}) in octahedral sites.



In inverse spinels, the divalent ions swap position with the trivalent ones, so that the former occupy only the octahedral sites and the latter also the tetrahedral ones. In random spinels, divalent and trivalent ions are divided into proportions 2:1 and 1:2 as shown in the figure.



The arrangement of spins in magnetite is such that the magnetic moments of Fe^{3+} ions, whose spin is $5/2$, cancel each other, leaving only the contribution of Fe^{2+} ions, whose spin is 2.

Magnetite is the progenitor of ferrites, with $\text{MO}\cdot\text{Fe}_2\text{O}_3$ structure, where M is a divalent cation between Fe, Zn, Cd, Ni, Cu, Co, Mg.

Exchange interactions in ferrimagnetic materials.

In ferrites, all exchange interactions favor antiparallel spin alignment. Since we have two magnetic centers A and B , it means that

$$J_{AA}, J_{AB}, J_{BB} < 0$$

However, if spins A are parallel to each other, same are spins B , then

$$|J_{AB}| > |J_{AA}| + |J_{BB}|$$

The condition for the stability of the spin arrangement in a ferrimagnet with permanent magnetization can be derived by the Weiss model.

Identify three types of exchange interactions, corresponding to interactions AA , BB and AB , characterized respectively by the molecular field coefficients $-\lambda$, $-\nu$ and $-\mu$, with a negative sign for what has just been said about the exchange integrals in ferrimagnets. According to Weiss' theory, spin A will be affected by exchange interactions with other spins A (with molecular field constant λ) but also with spin B (with molecular field constant μ), and vice versa for B (with molecular field constants ν and μ). Then the Weiss field heard by spins A and B will be:

$$\bar{B}_A = -\lambda \bar{M}_A - \mu \bar{M}_B$$

$$\bar{B}_B = -\nu \bar{M}_B - \mu \bar{M}_A$$

They will correspond to an interaction energy density²⁷:

$$U = -\frac{I}{2}(\bar{B}_A \cdot \bar{M}_A + \bar{B}_B \cdot \bar{M}_B) = \frac{I}{2}\lambda M_A^2 + \mu \bar{M}_A \cdot \bar{M}_B + \frac{I}{2}\nu M_B^2$$

which is less than 0 when $\bar{M}_A$ it is antiparallel to $\bar{M}_B$ and the interaction AB prevails over the sum of the interactions AA and BB :

$$\bar{M}_A \cdot \bar{M}_B < 0$$

$$\mu M_A M_B > \frac{I}{2}(\lambda M_A^2 + \nu M_B^2)$$

as in the case of ferrites, where the ground state has MA antiparallel to MB and the material is ferrimagnetic.

Susceptibility of ferrimagnetic materials.

To derive the susceptibility of ferrimagnets, two Curie constants can be defined for spins A and B . For simplicity, consider only the interaction AB (which has just been seen to be the prevailing one):

$$\bar{B}_A = -\mu \bar{M}_B$$

$$\bar{B}_B = -\mu \bar{M}_A$$

²⁷ If magnetization per unit volume is used, the energy will be per unit volume.

According to the Weiss model one can write²⁸

$$\left. \begin{aligned} \frac{M_A}{B} = \chi_A = \frac{C_A}{T} \\ B_A = -\mu M_B \\ \frac{M_B}{B} = \chi_B = \frac{C_B}{T} \\ B_B = -\mu M_A \end{aligned} \right\} \rightarrow \left. \begin{aligned} M_A &= \frac{C_A}{T} (B_a - \mu M_B) \\ M_B &= \frac{C_B}{T} (B_a - \mu M_A) \end{aligned} \right\}$$

To find the Curie temperature I can solve the system of two equations in case $B_a = 0$:

$$\left. \begin{aligned} TM_A + C_A \mu M_B &= 0 \\ TM_B + C_B \mu M_A &= 0 \end{aligned} \right\} \rightarrow \begin{vmatrix} T & C_A \mu \\ C_B \mu & T \end{vmatrix} = 0 \rightarrow T^2 - C_A C_B \mu^2 = 0$$

from which one derives T_c :

$$T_c = \mu (C_A C_B)^{1/2}$$

which is the only temperature at which you can have M_A and M_B other than 0 even in the absence of B_a (for $T > T_c$ and $B_a = 0$, the system admits only the trivial solution $M_A = M_B = 0$).

Solving the system for $B_a \neq 0$ one gets M_A and M_B :

$$\left. \begin{aligned} M_A &= \frac{C_A}{T} (B_a - \mu M_B) \\ M_B &= \frac{C_B}{T} \left(B_a - \mu \frac{C_A}{T} (B_a - \mu M_B) \right) \end{aligned} \right\} \rightarrow \left. \begin{aligned} M_A &= \frac{C_A}{T} (B_a - \mu M_B) \\ M_B &= \frac{C_B B_a}{T} - \mu \frac{C_B C_A}{T^2} B_a + \mu^2 \frac{C_B C_A}{T^2} M_B \end{aligned} \right\} \rightarrow$$

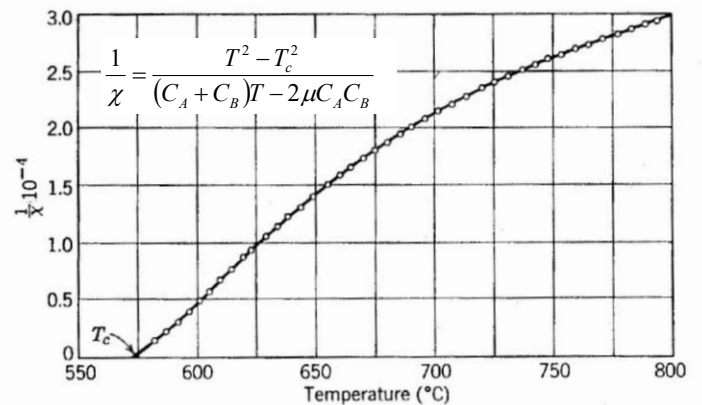
$$\left. \begin{aligned} M_A &= \frac{C_A}{T} (B_a - \mu M_B) \\ M_B &= \frac{C_B T - \mu C_B C_A}{T^2} B_a \frac{T^2}{T^2 - \mu^2 C_B C_A} \end{aligned} \right\} \rightarrow \left. \begin{aligned} M_A &= \frac{C_A}{T} \left(B_a - \mu \frac{C_B T B_a - \mu C_B C_A B_a}{T^2 - \mu^2 C_B C_A} \right) \\ M_B &= \frac{C_B T B_a - \mu C_B C_A B_a}{T^2 - \mu^2 C_B C_A} \end{aligned} \right\} \rightarrow$$

$$\left. \begin{aligned} M_A &= \frac{C_A T B_a - \mu C_B C_A B_a}{T^2 - \mu^2 C_B C_A} \\ M_B &= \frac{C_B T B_a - \mu C_B C_A B_a}{T^2 - \mu^2 C_B C_A} \end{aligned} \right\}$$

that substituting in susceptibility leads to

$$\chi = \frac{M_A + M_B}{B_a} = \frac{(C_A + C_B)T - 2\mu C_A C_B}{T^2 - T_c^2} \quad (\text{c.g.s.})$$

The figure on the side shows the reciprocal susceptibility of magnetite as a function of temperature. We observe the typical trend of



²⁸ It is also assumed in this case that $B = \mu_0 H + \mu_0 M \approx \mu_0 H$ since $\chi_m \ll 1 \rightarrow \mu_r \sim 1$ and M is very small.

ferrimagnets, which is different from that of the Curie-Weiss law (for $T > T_c$ is

$$\chi \propto \frac{C}{T - T_c}, \text{ and for } T \sim T_c \text{ is } \chi \propto \frac{C}{(T - T_c)^\gamma} \text{ followed by ferromagnets.}$$

Anti-ferromagnetism.

The coupling of atomic magnetic moments does not always lead to a "constructive" alignment as in ferromagnetism.

For some materials, the alignment of the spin moments of adjacent atoms occurs in opposite directions.

For example, in MnO, O^{2-} has no net magnetic moment, while Mn^{2+} has a net moment due to the spin of electrons in the d orbitals. However, below a certain temperature, there is no net magnetization in the material, so MnO is called antiferromagnetic.

In an antiferromagnetic material, below a magnetic ordering temperature called the Néel T_N temperature, the spins are arranged in an antiparallel manner.

Antiferromagnetism represents a special case of ferrimagnetism in which there are two spin sub-crystals with

equal and opposite magnetization. Thus, it is equivalent to placing $C_A = C_B$ in the susceptibility in the paramagnetic range:²⁹

$$\chi = \frac{M_A + M_B}{B_a} = \frac{(C_A + C_B)T - 2\mu C_A C_B}{T^2 - T_c^2} = \frac{2CT - 2C^2\mu}{T^2 - T_c^2} \text{ (c.g.s.)}$$

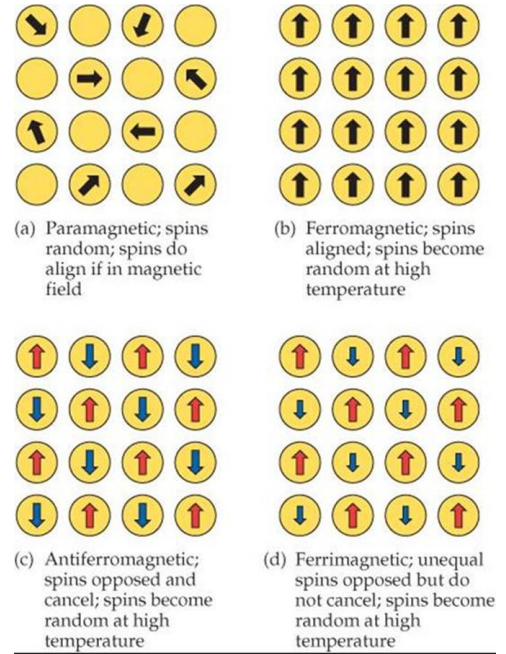
with

$$T_c = \mu(C_A C_B)^{1/2} = \mu C = T_N$$

whereby

$$\begin{aligned} \chi &= \frac{2CT - 2C^2\mu}{T^2 - \mu^2 C^2} = 2C \frac{T - \mu C}{(T + \mu C)(T - \mu C)} = \frac{2C}{T + \mu C} \\ &= \frac{2C}{T + T_N} \end{aligned}$$

In fact, experimentally we find a trend of the type



²⁹ It is also assumed in this case that $B = \mu_0 H + \mu_0 M \approx \mu_0 H$ since $\chi_m \ll 1 \rightarrow \mu_r \sim 1$ and M is very small.

$$\chi = \frac{2C}{T + \theta}$$

where θ is a fictitious temperature not coinciding with T_N at which the transition from para to antiferromagnet is actually observed, in fact θ/T_N varies from ~ 5 to ~ 1 . The discrepancy is due to the need to also consider the exchange interactions with the second closest primes and the possible arrangement in sublattices other than the crystalline one.

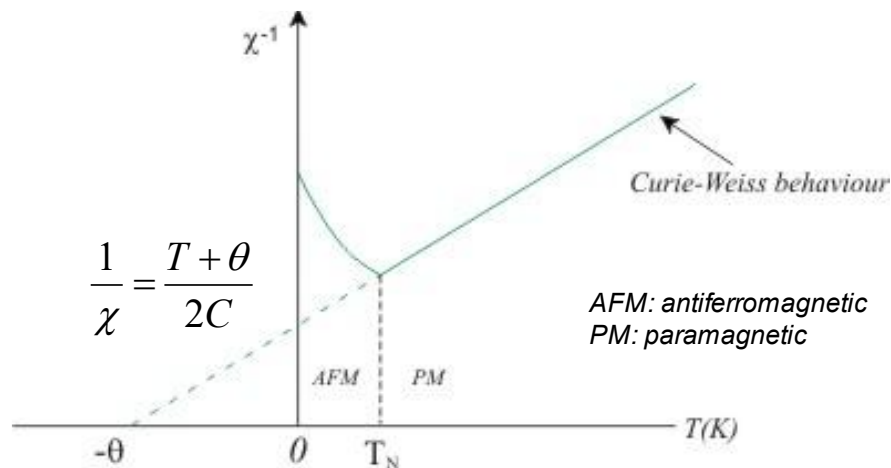


Table 3 Antiferromagnetic crystals

Substance	Paramagnetic ion lattice	Transition temperature, T_N , in K	Curie-Weiss θ , in K	$\frac{\theta}{T_N}$	$\frac{\chi(0)}{\chi(T_N)}$
MnO	fcc	116	610	5.3	$\frac{2}{3}$
MnS	fcc	160	528	3.3	0.82
MnTe	hex. layer	307	690	2.25	
MnF ₂	bc tetr	67	82	1.24	0.76
FeF ₂	bc tetr	79	117	1.48	0.72
FeCl ₂	hex. layer	24	48	2.0	<0.2
FeO	fcc	198	570	2.9	0.8
CoCl ₂	hex. layer	25	38.1	1.53	
CoO	fcc	291	330	1.14	
NiCl ₂	hex. layer	50	68.2	1.37	
NiO	fcc	525	~ 2000	~ 4	
Cr	bcc	308			

Susceptibility of antiferromagnets.

While above T_N (in the paramagnetic phase) the susceptibility is independent of the direction of application of the magnetizing field, below T_N there are two possible "limit" orientations: parallel to the spins or perpendicular to them.

In general, the energy density in the presence of a magnetic field will be given by the balance between two contributions, the one due to the tendency of the two magnetizations to arrange

themselves in an antiparallel direction, and the opposite one due to the tendency of the magnetizations to align in the direction of the magnetizing field:

$$U = \mu \bar{M}_A \cdot \bar{M}_B - \bar{B}_a \cdot \bar{M}_A - \bar{B}_a \cdot \bar{M}_B = \mu \bar{M}_A \cdot \bar{M}_B - B_a M_A \cos \theta_A - B_a M_B \cos \theta_B$$

where θ_i is the angle between the field and the magnetization.

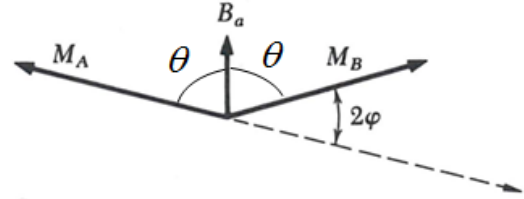
When the field is perpendicular to the direction of magnetization, you will have

$$\theta_A = \theta_B = \theta$$

So what

$$\bar{M}_A = -\bar{M}_B \rightarrow |\bar{M}_A| = |\bar{M}_B| = M$$

and the two magnetizations will be distorted in the direction of the field itself due to the tendency of the spins to align with the magnetizing field (see figure), so an angle 2ϕ will be formed between them



$$2\theta + 2\phi = \pi \rightarrow \theta = \frac{\pi - 2\phi}{2}$$

so you can write

$$U = \mu M^2 \cos(\pi - 2\phi) - 2B_a M \cos\left(\frac{\pi - 2\phi}{2}\right)$$

Since

$$\cos(x - y) = \cos(x)\cos(y) + \sin(y)\sin(x)$$

one has

$$\cos \theta = \cos \frac{\pi - 2\phi}{2} = \cos \frac{\pi}{2} \cos \phi + \sin \frac{\pi}{2} \sin \phi = 0 + \sin \phi$$

$$\cos(\pi - 2\phi) = \cos \pi \cos 2\phi + \sin \pi \sin 2\phi = -\cos 2\phi + 0$$

Being 2ϕ very small, it can expand in power series truncating to first order in ϕ

$$\cos 2\phi \approx 1 - \frac{1}{2}(2\phi)^2 + \dots$$

$$\sin \phi \approx \phi + \dots$$

so one rewrites U as

$$U \cong \mu M^2 \left(-1 + \frac{1}{2}(2\phi)^2 \right) - 2B_a M \phi$$

Then it was possible to express the magnetic energy as a function of the angle 2ϕ and its value corresponding to the minimum of U can be identified:

$$\frac{dU}{d\phi} = 4\mu M^2 \phi - 2B_a M = 0 \rightarrow \phi = \frac{B_a}{2\mu M}$$

So the resulting magnetization in the material will be

$$M_{tot} = 2M \cos \theta = 2M \sin \phi \approx 2M\phi =$$

$$= 2M \frac{B_a}{2\mu M} = \frac{B_a}{\mu}$$

and susceptibility is by definition

$$\chi = \frac{M_{tot}}{B_a}$$

and the magnetic susceptibility is obtained by field oriented perpendicular to the direction of the two magnetizations:

$$\chi_{\perp} = \frac{1}{B_a} \frac{B_a}{\mu} = \frac{1}{\mu}$$

In the other extreme case of B_a parallel to the direction of the magnetizations, if all spins are aligned and equally distributed between M_A and M_B (i.e. at 0 K), then

$$M_{tot} = M_A - M_B = 0$$

and

$$\chi_{\parallel} = \frac{M_{tot}}{B_a} = 0.$$

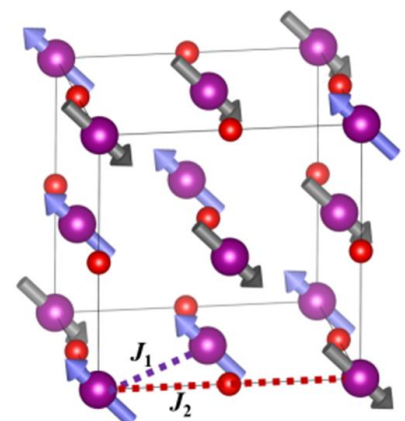
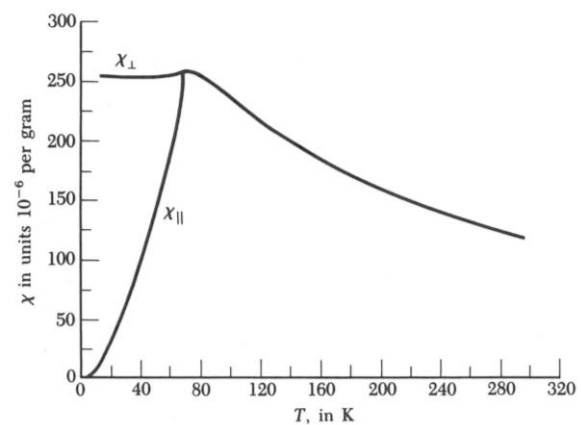
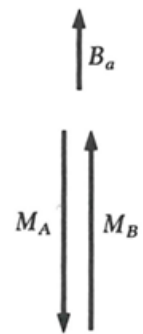
La $\chi_{\parallel}$ will progressively tend to the value of T_N susceptibility due to spin misalignment due to thermal agitation.

Overall, a material will have a series of domains with various orientations and the resulting susceptibility at the macroscopic level will be the average of the two.

$$\chi = \frac{1}{3} \chi_{\perp} + \frac{2}{3} \chi_{\parallel}$$

so, we see that at T_N susceptibility has a cusp, instead of diverging as in ferromagnetic materials.

The magnetic structure of MnO is shown in the figure on the side, where Mn^{2+} ions are indicated in purple and O^{2-} ions in red. It is observed that, along the side of the cell, the distance between the Mn^{2+} ions is greater than that which usually admits an intense



exchange interaction. In this case, antiferromagnetism takes place due to the mediation of the oxygen atom placed between the two ions of Mn^{2+} , based on a mechanism called "super exchange". In the "super exchange", the d electrons of Mn are coupled with the electrons of O, which in turn interact with the d electrons of the other ion of Mn.

The dashed line corresponds to the direct exchange interaction J_1 between adjacent Mn ions (near primes), while the dashed line J_2 shows a "super-exchange" interaction with the second closest primes, through an intermediate O ion. The result is that, below T_N , the spins arranged in the single plane (111) are parallel to each other, but the direction of the spins of two adjacent planes (111) is antiparallel.

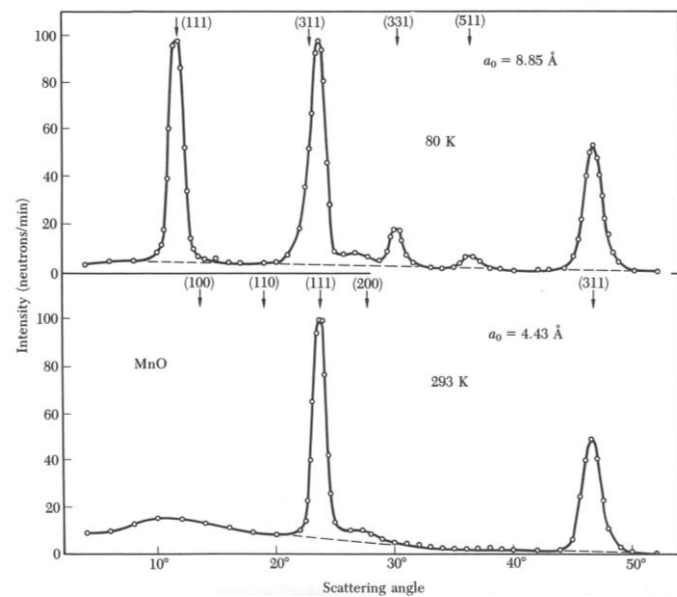
Neutron scattering.

The phase transition from paramagnetic to antiferromagnetic can be measured through neutron scattering. In fact, neutrons, despite having no charge, have spin, which allows them to interact with the spin "lattice" of the material, be it electronic or nuclear. The magnetic moment of neutrons interacts with those present in the material, and the neutron scattering cross section is of the same order of magnitude whether the scattering occurs with electronic magnetic moments, or with nuclear ones ($\sim 10^3$ times smaller), indicating that the interaction with nuclei is $\sim 10^3$ times more intense than with electrons. Neutrons interact with magnetic moments in all their forms, and in fact can lead to annihilation or excitation of a magnon, through an inelastic scattering process. In this way it is possible to measure the characteristic frequency of magnons ω_m according to the principle of conservation of energy:

$$\frac{\hbar^2 k_{n,i}^2}{2M_n} = \frac{\hbar^2 k_{n,s}^2}{2M_n} + \hbar\omega_m$$

where $k_{n,i}$ and $k_{n,s}$ are respectively the wave vector of the incident and scattered neutron, whose mass is M_N .

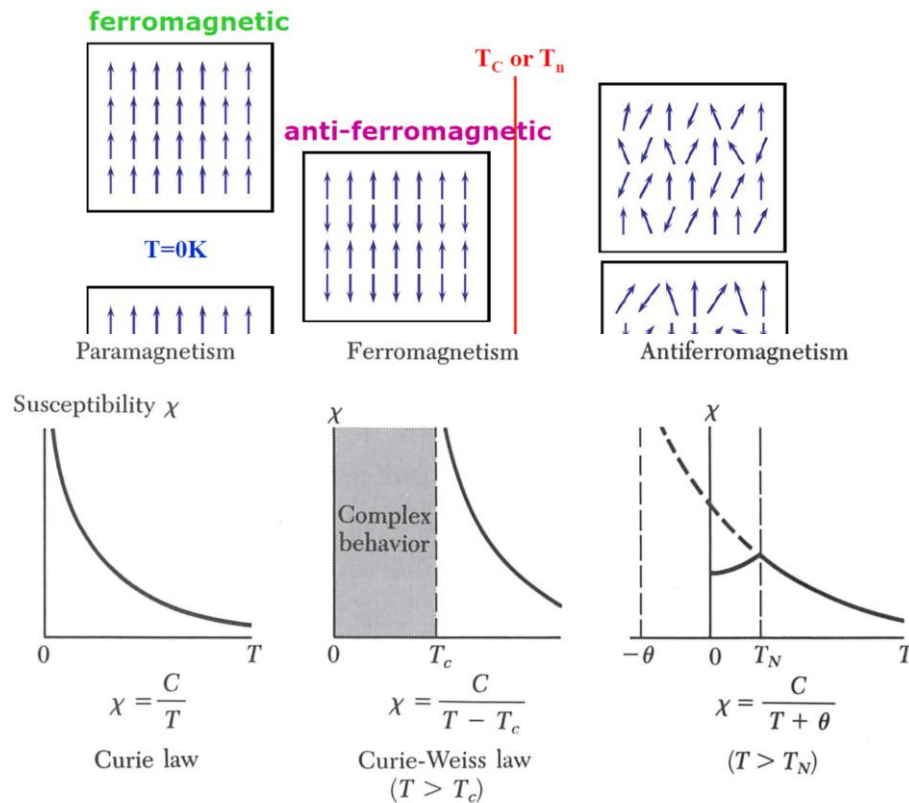
The neutron scattering measurement performed on MnO, at $T > T_N = 120 \text{ K}$ shows peaks of scattering intensity coinciding with those of a cubic cell F.C.C. with cell parameter $a = 4.43 \text{ \AA}$, while for $T < T_N$ the spin cell is cubic with cell parameter $a = 8.85 \text{ \AA}$. Since no structural phase transition occurs in the material, both at $T < T_N$ and at



$T > T_N$ the crystalline cell always remains the cubic F.C.C. with cell parameter $a = 4.43 \text{ \AA}$.

Dependence of susceptibility on temperature.

Above the critical (or Curie) temperature, ferro- and ferrimagnetic materials no longer possess spontaneous magnetization, becoming paramagnetic. The same thing happens for antiferromagnets.



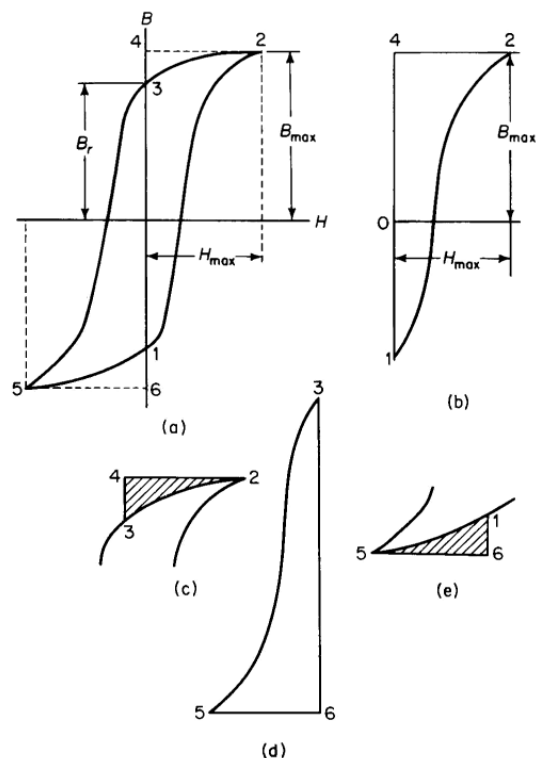
Applications of iron and ferrimagnetic materials.

The shape of the hysteresis cycle reflects the magnetic anisotropy of the material and determines the applications of the ferromagnet.

In fact, the energy expended during the hysteresis cycle is proportional to the area of the hysteresis cycle itself, and for a material of volume V is given by

$$E = V \oint H dB$$

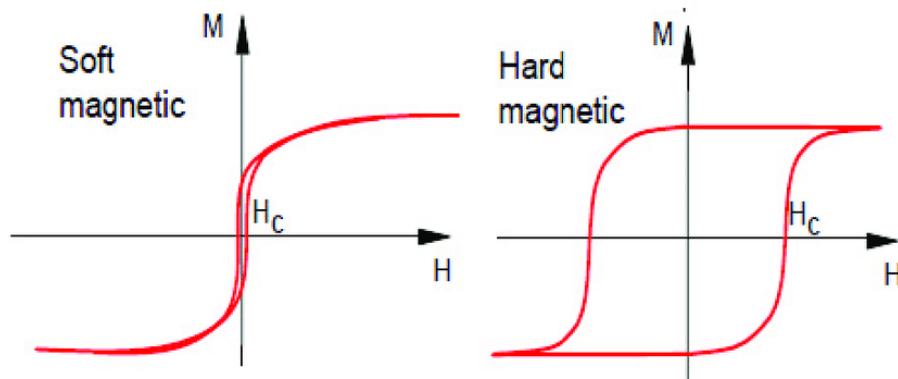
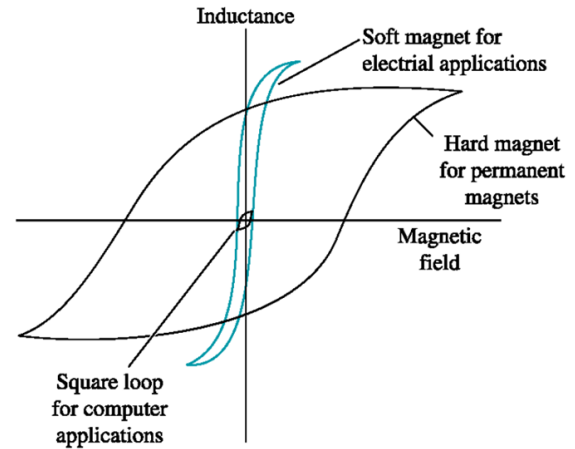
As shown in the figure, sections 1-2 and 3-5 make a positive contribution to the work done for magnetization (in 1-2 $H > 0$ and $dB > 0$, in 3-5 $H < 0$ and $dB < 0$), while sections 2-3 and 5-6 give a



negative contribution (in 2-3 $H > 0$ and $dB < 0$, in 5-6 $H < 0$ and $dB > 0$).

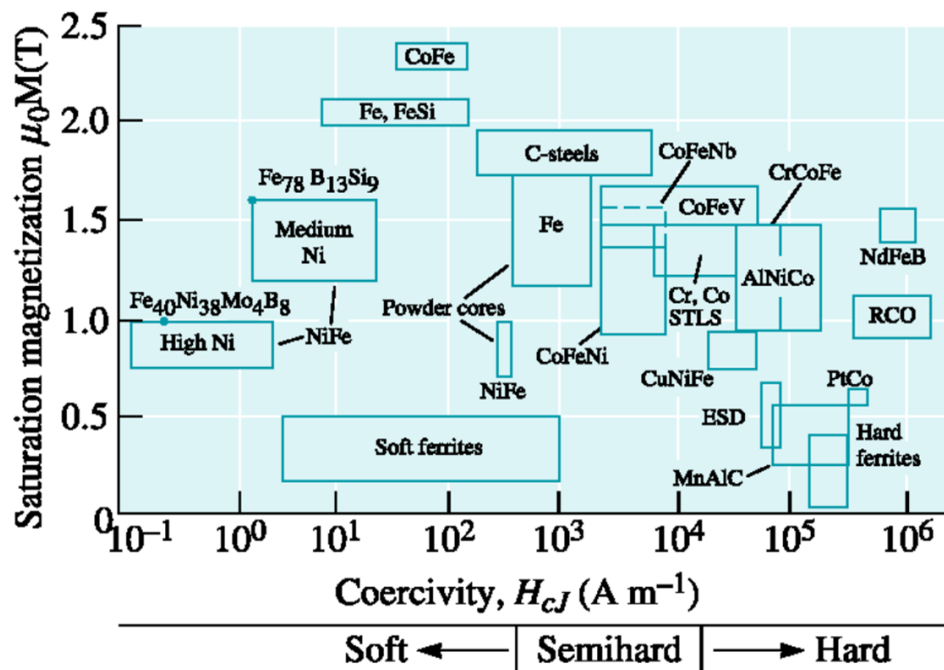
The figure shows the comparison between hysteresis cycles for three different applications of magnetic materials:

- "Soft" magnets, in which the coercive field is relatively low compared to magnetization.
- "Hard" magnets, in which the coercive field is high.
- Magnets with "square" cycle, in which the residual magnetization is comparable to that of saturation and the coercive field is relatively high.



Types of magnetic materials.

The graph below represents the saturation magnetization and coercive field for a range of magnetic materials.



Soft Ferromagnets:

- so-called "electric" steels for applications in electronic devices. They are bonded with Si to increase resistivity and decrease magnetic anisotropy
- Fe-Co-Mn
- permalloys ($\text{Ni}_{78}\text{Fe}_{22}$) with almost zero anisotropy and magnetostriction
- vitreous metals (e.g. $\text{Fe}_{79}\text{B}_{13}\text{Si}_9$ based)

Hard Ferromagnets:

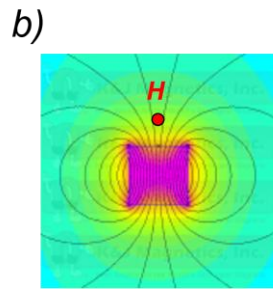
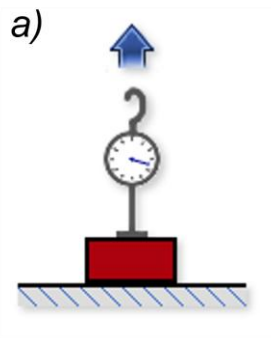
- Rare earth alloys with Mn, Fe, Co and Ni, such as SmCo_5 , $\text{Nd}_2\text{Fe}_{14}\text{B}$.

Type of ferromagnet	H_c	Application
Alnico V	600 g	Speakers
SmCo_5	10000 g	Special magnets with high stability
Fe 4wt% in Si	0.5 g	Transformers
Supermalloy	0.002 g	Pulsed transformers

Classification of magnets

In the industrial or application field, magnets can be evaluated according to different parameters such as:

- the tensile force, i.e. the force to be exerted to separate a magnet from an iron surface (with standard characteristics). It can be expressed in N or Kg .
- the strength of the magnetic field, measured at a certain distance and direction from the surface of the magnet. Expressed in *Gauss* or T .
- the maximum "energy product", which corresponds to the maximum value of the product $H*B$ in the hysteresis cycle H vs B . This value, measured in Mega Gauss Oersted ($MGOe$) is typically anticipated by the letter N and can be followed by other letters (M , H , SH , UH , EH , AH) that indicate the maximum operating temperature without loss of magnetization (for example $N45M$ means $45 MGOe$ and maximum operating temperature of 100°C).



c)

Magnet Type	Max Energy Product (MGOe)
Neodymium	35-52
SmCo 26	26
Alnico 5/8	5.4
Ceramic	3.4
Flexible	0.6-1.2

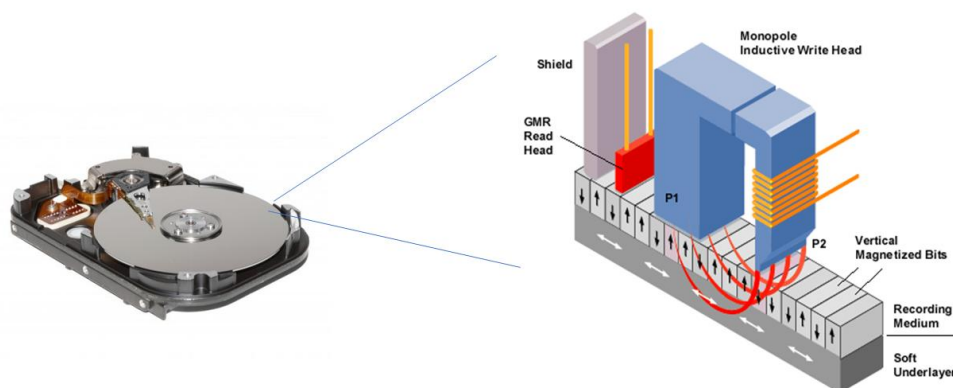
As for applications:

- Soft magnetic materials are often used to amplify the magnetic flux density produced by a coil through which electric current passes. This series includes electromagnets, electric motors, transformers, generators, and other electrical equipment. In these electrical devices it must be easy to cancel the induced magnetization without dissipating an excessive amount of heat. For the core of a transformer you want a ferromagnet with low coercive field, because this corresponds to low dissipation of magnetic energy into thermal energy at each hysteresis cycle.

In the particular case of ferrites, they are oxides and, therefore, are not electrical conductors, which allows them to be used as dielectric magnetic components, in high-frequency applications such as microwave devices (e.g. circulators and phase modulators).

- Hard magnetic materials are used as permanent magnets. Permanent magnets can be classified according to the "power" of the magnet, i.e. the maximum value reached by the product of magnetic induction for the magnetizing field inside the hysteresis (rectangle of greater area inscribable in hysteresis), for which high coercivity is needed.

- Square hysteresis materials are used for data storage such as hard drives and magnetic tape devices such as cassette tapes or advanced tape storage systems. In fact, they maintain a high magnetic field even after the removal of the magnetizing field.



Permanent magnet and induction motors.

Induction motors use ferromagnetic materials, mainly silicon-steel laminations, in both stator and rotor cores. These materials have high magnetic permeability, which channels and strengthens the magnetic flux. The laminations reduce eddy current losses, improving efficiency. Copper or aluminum conductors are embedded in the rotor to carry induced currents. The interaction between the magnetic core fields and rotor currents produces torque. Thus, magnetic materials are essential for flux concentration, efficiency, and torque generation.

The reverse mechanism is exploited for electricity generation. Here, rotation due to a mechanical force drives the rotor. This changing magnetic flux induces voltage in the stator windings. Thus, mechanical energy is converted into electrical energy by the same principles.

In permanent magnet motors, the rotor carries magnets that establish a steady magnetic field. This field interacts with the stator's current-carrying windings to produce torque. Since no field windings are required, copper losses in excitation are avoided. The design offers high efficiency and power density compared to induction motors. Stronger magnets, such as NdFeB, allow compact machines with high torque output. Precise control is achieved with modern power electronics and sensors. These motors are widely used in electric vehicles, wind turbines, and robotics. Their combination of efficiency, compactness, and reliability makes them vital in advanced systems.

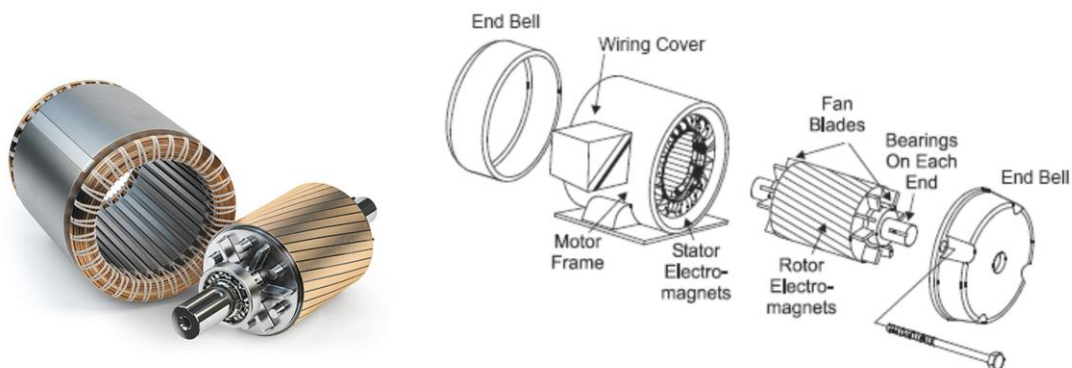


Figure 3-28. Induction motor construction

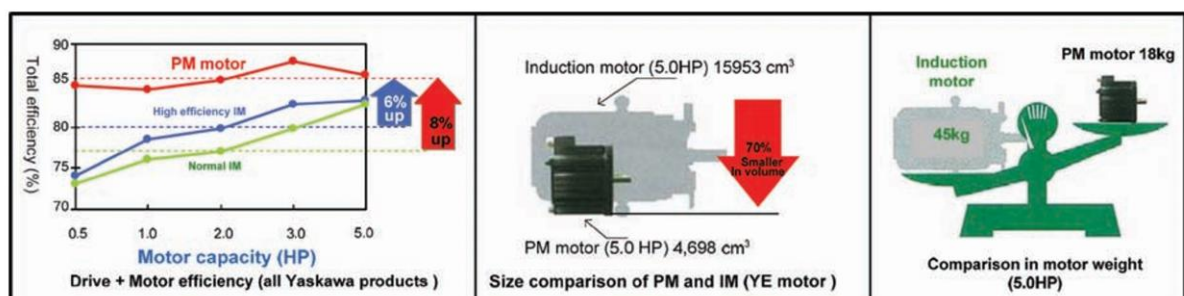
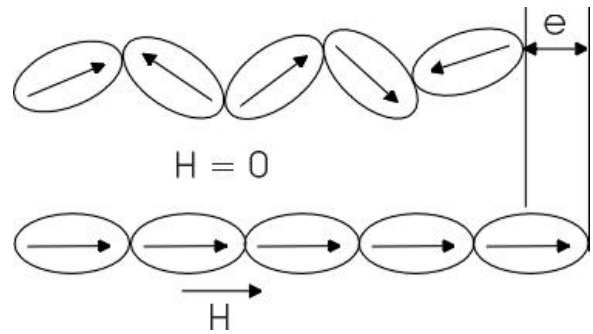


Figure 8. Some advantages of a PM motor over an induction motor (IM). Reproduced with permission.^[33]

Magnetostriction.

Magnetostriction is the variation in the size of a material, usually ferromagnetic metallic, due to its magnetization. This phenomenon is due to the reorientation of atomic dipoles. Since the process is isochoric and takes place in the direction of magnetization, it actually consists in the elongation of



the material in the direction of the magnetizing field. This effect is in some ways equivalent to the piezoelectric effect, and in the past it was used to generate sound waves (e.g. sonar during World War 2).

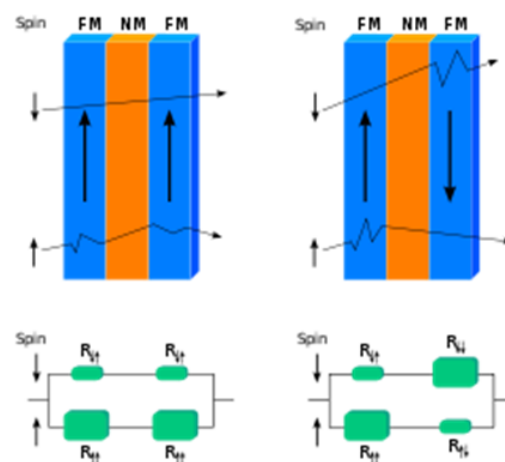
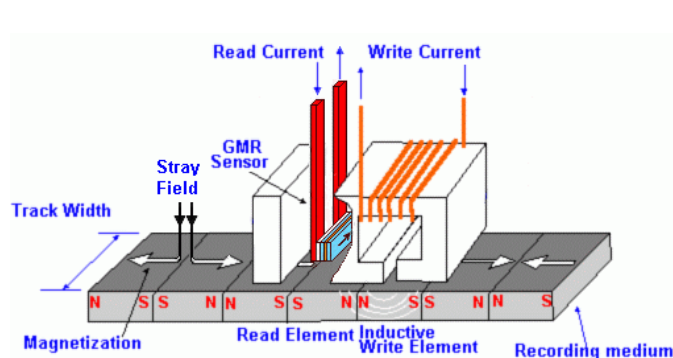
Giant magnetoresistance.

Giant magnetoresistance (GMR) is a quantum effect of electrical resistance dependent on the magnetization of the material, observed in thin-film structures composed of alternating layers of ferromagnetic and non-magnetic type. The 2007 Nobel Prize in Physics was awarded to Albert Fert and Peter Grünberg for the discovery of GMR.

The effect is observed as a significant change in electrical resistance depending on whether the magnetization of adjacent ferromagnetic layers is in parallel or antiparallel alignment. The overall resistance is relatively low for parallel alignment and relatively high for antiparallel alignment.

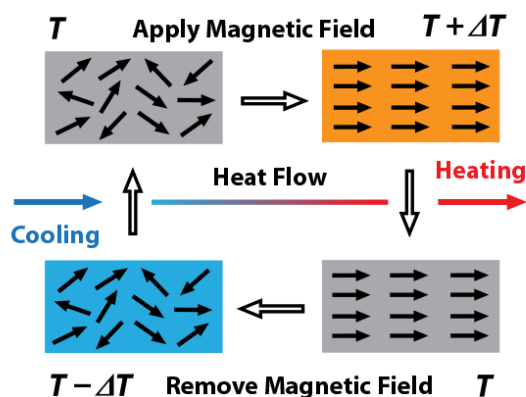
The typical GMR-based device is the spin-valve. The figure below shows a multilayer composed of a non-magnetic conducting film (NM) coupled to two ferromagnetic (FM) conducting films. You can have the configuration with antiparallel (right) and parallel (left) magnetizations. The vertical black arrows in the ferromagnetic layers show the direction of magnetization, while the arrows through the spin valve schematize the path of the electron, where a deviation from the straight path indicates that the electron has been scattered. In the spin valve, an electron passing through the FM layer will suffer a higher probability of scattering if it has spin opposite to the direction of magnetization. This principle allows the two FM layers to be represented as two electrical circuits in parallel, one for each spin orientation of the conduction electrons, where the resistance is different depending on the orientation of the spin. The calculation of the equivalent resistance for the spin valve in antiparallel and parallel configuration shows that parallel alignment has the lowest resistance.

GMR is used in hard drives to read stored data.



Magnetocaloric effect and magnetic refrigeration.

The magnetocaloric effect consists of heating a magnetic material following its exposure to an external magnetic field. The increase in temperature is due to the heat released into the material as a result of the spin being sorted along the direction of the external field. The magnetocaloric effect is more intense close to the Curie temperature of the material. In some cases, magnetic sorting also involves a structural change, which corresponds to an additional heat release.



The magnetocaloric effect is used in magnetic refrigeration, which allows to obtain extremely low temperatures, useful for scientific purposes, but above all as a refrigeration technology for common refrigerators.

The principle of magnetic refrigeration is that, after the magnetic ordering, the ferromagnet will return to thermal equilibrium with the environment. At this point, if the magnetizing field is removed, the ferromagnet will be able to absorb heat from the outside, which it will use to return the magnetization to its original value. During this process, the ferromagnet acts as a chiller. By coupling the ferromagnet with a fluid heat exchange system (e.g. water), the heat released during the magnetization cycle can be dissipated outwards, and the heat inside the luminaire can be absorbed during the demagnetization cycle.

Compared to traditional gas compression refrigeration, magnetic refrigeration is quieter, more compact, has superior cooling efficiency and does not use harmful gases for the Earth's ozone layer.